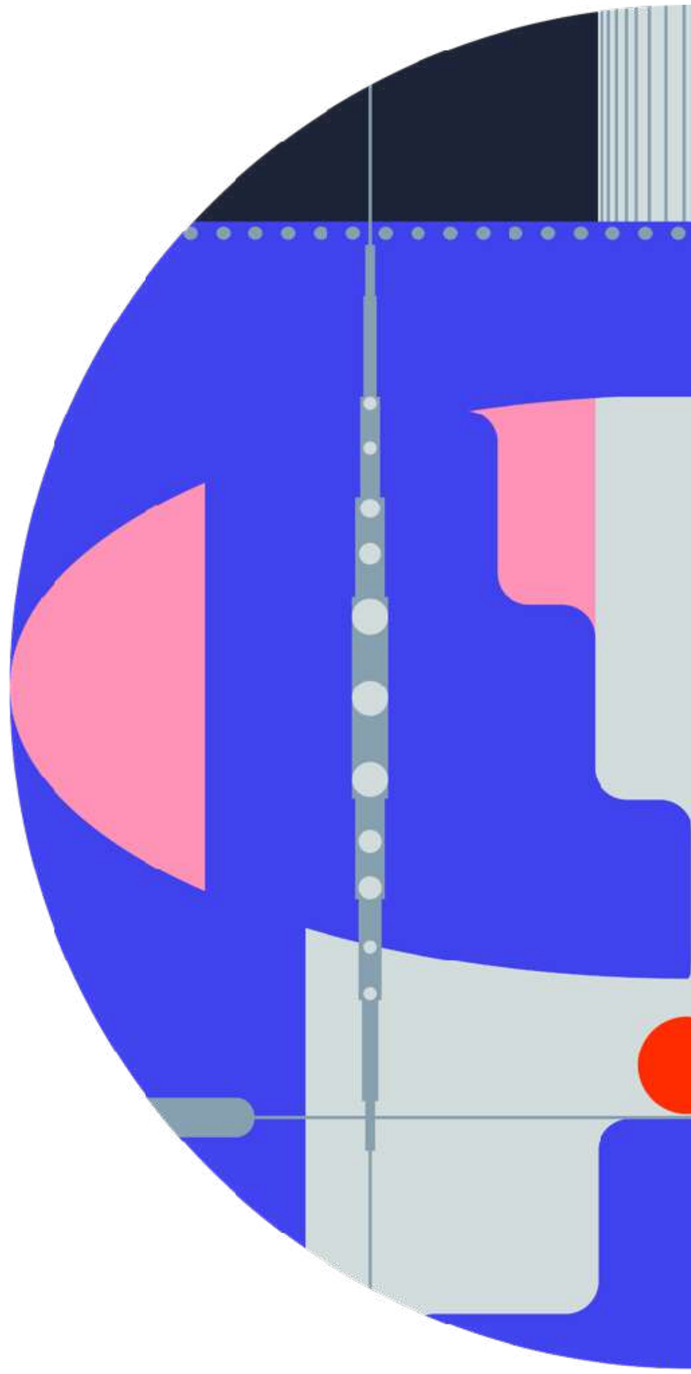
Diffusion models in practice



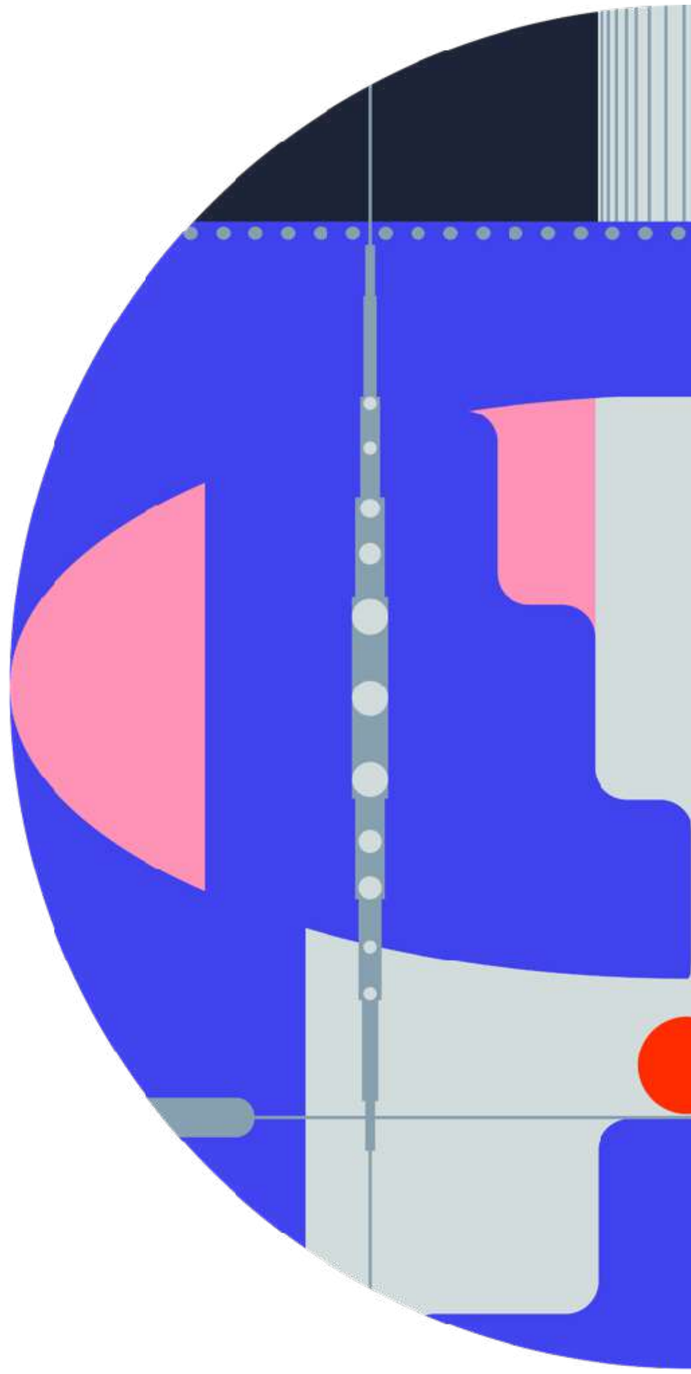
Outline

01 “Quick” recap

02 Designing and training diffusion models

03 Advanced sampling techniques in diffusion models

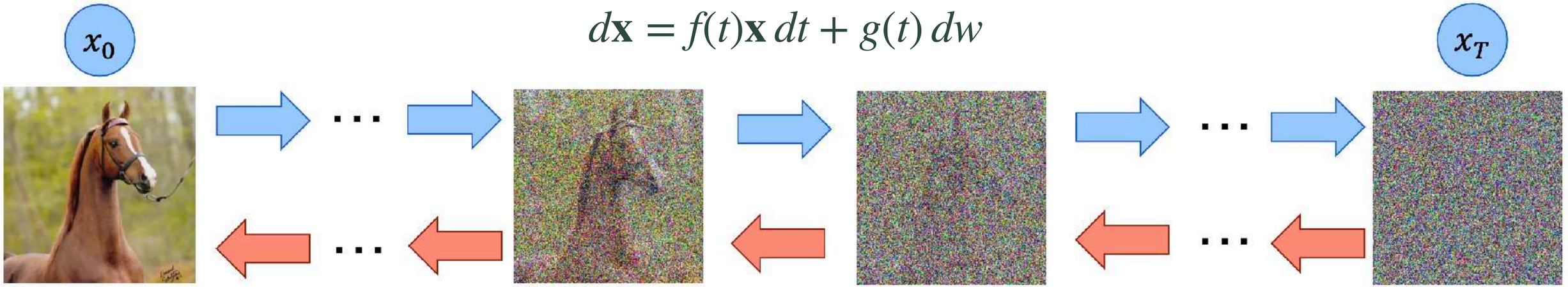
“Quick” recap on diffusion models



Stochastic differential equations

Forward SDE

$$d\mathbf{x} = f(t)\mathbf{x}\,dt + g(t)\,dw$$



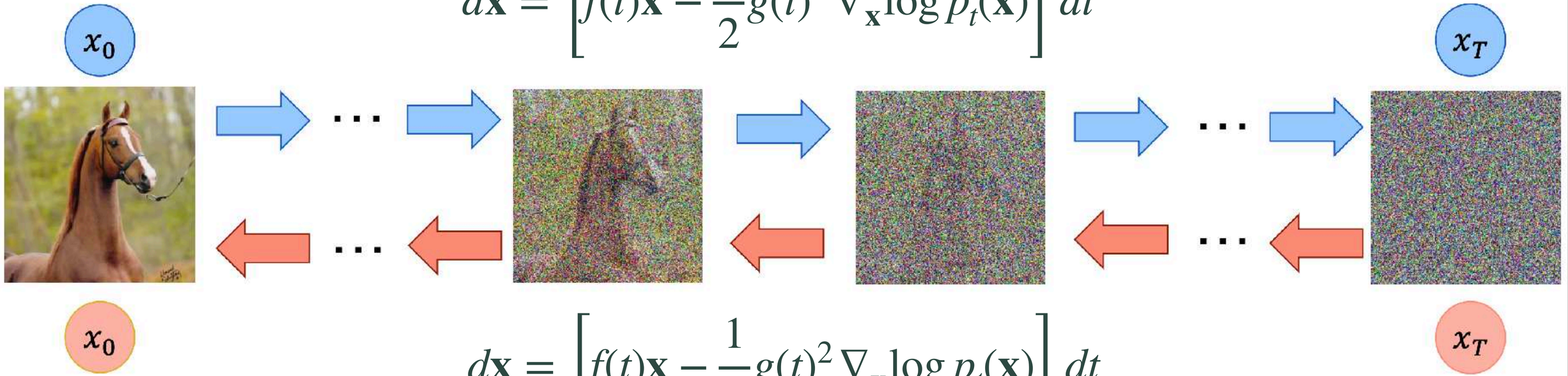
$$d\mathbf{x} = \left[f(t)\mathbf{x} - g(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x})\right] dt + g(t)\,d\bar{w}$$

Reserve SDE

Y. Song, et al. Score-based generative modeling through stochastic differential equations. 2021

Probability Flow ODE

$$d\mathbf{x} = \left[f(t)\mathbf{x} - \frac{1}{2} g(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x}) \right] dt$$



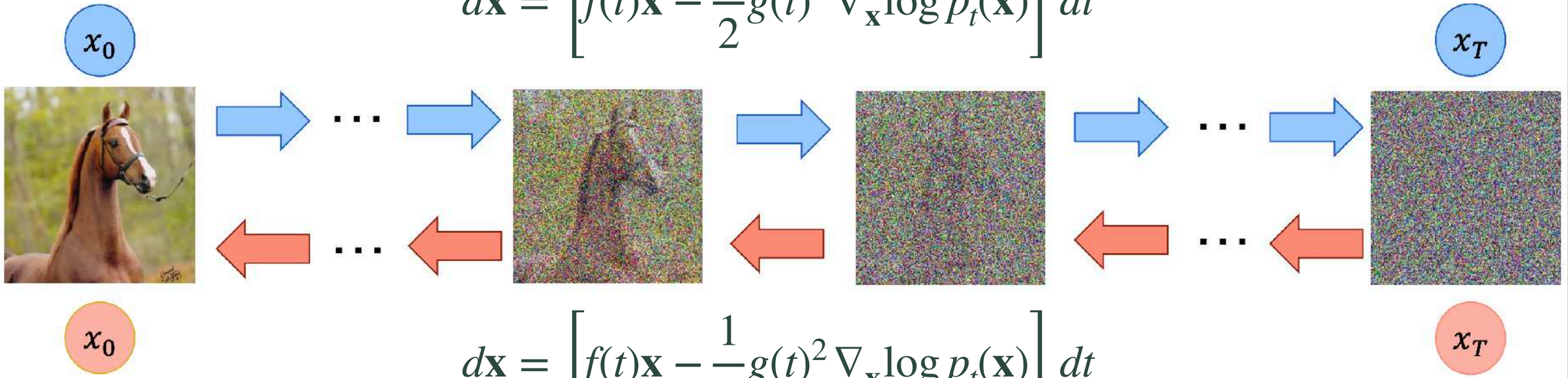
$$d\mathbf{x} = \left[f(t)\mathbf{x} - \frac{1}{2} g(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x}) \right] dt$$

Y. Song, et al. Score-based generative modeling through stochastic differential equations. 2021

Flow matching

$$\mathbf{v}(\mathbf{x}, t)$$

$$d\mathbf{x} = \left[f(t)\mathbf{x} - \frac{1}{2} g(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x}) \right] dt$$



$$d\mathbf{x} = \left[f(t)\mathbf{x} - \frac{1}{2} g(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x}) \right] dt$$

$$\mathbf{v}(\mathbf{x}, t)$$

Y. Lipman, et al. Flow Matching for Generative Modeling. 2022

Forward process

General Forward Process: $p(\mathbf{x}_t \mid \mathbf{x}_0) = \mathcal{N}(\alpha_t \mathbf{x}_0, \sigma_t^2 I)$

$$\mathbf{x}_t = \alpha_t \mathbf{x}_0 + \sigma_t \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

Variance Preserving (VP): $\alpha_t^2 + \sigma_t^2 = 1$

DDPM linear-β: $\alpha_t = e^{-\frac{1}{2}\int_0^t \beta(s)\, ds}$ **Cosine:** $\alpha_t = \cos\left(\frac{\pi t}{2}\right), \quad \sigma_t = \sin\left(\frac{\pi t}{2}\right)$

Variance Exploding (VE): $\alpha_t = 1, \quad \sigma_t = t$

Linear Interpolation (Flow Matching): $\alpha_t = 1 - t, \quad \sigma_t = t$

Training

Score Matching

Tweedie's formula

$$\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t) = \nabla_{\mathbf{x}_t} \log \int p_{data}(\mathbf{x}) p(\mathbf{x}_t \mid \mathbf{x}) d\mathbf{x} = \boxed{\mathbb{E}\left[\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t \mid \mathbf{x}_0) \mid \mathbf{x}_t\right]}$$

Intractable target

$$\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t \mid \mathbf{x}_0) = -\frac{1}{\sigma_t^2}\left(\mathbf{x}_t - \alpha_t \mathbf{x}_0\right) = -\frac{\boldsymbol{\epsilon}}{\sigma_t}$$

Conditional trick

$$\mathop{\mathbb{E}}_{t\sim p(t)} \mathop{\mathbb{E}}_{\mathbf{x}_t \sim p(\mathbf{x}_t)} \left[w(t) \cdot \|\mathbf{s}_\theta(\mathbf{x}_t, t) - \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t)\|_2^2\right] = \mathop{\mathbb{E}}_{t\sim p(t)} \mathop{\mathbb{E}}_{\mathbf{x}_t \sim p(\mathbf{x}_t)} \left[w(t) \cdot \|\mathbf{s}_\theta(\mathbf{x}_t, t) - \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t \mid \mathbf{x}_0)\|_2^2\right] + C$$

Training

Flow Matching

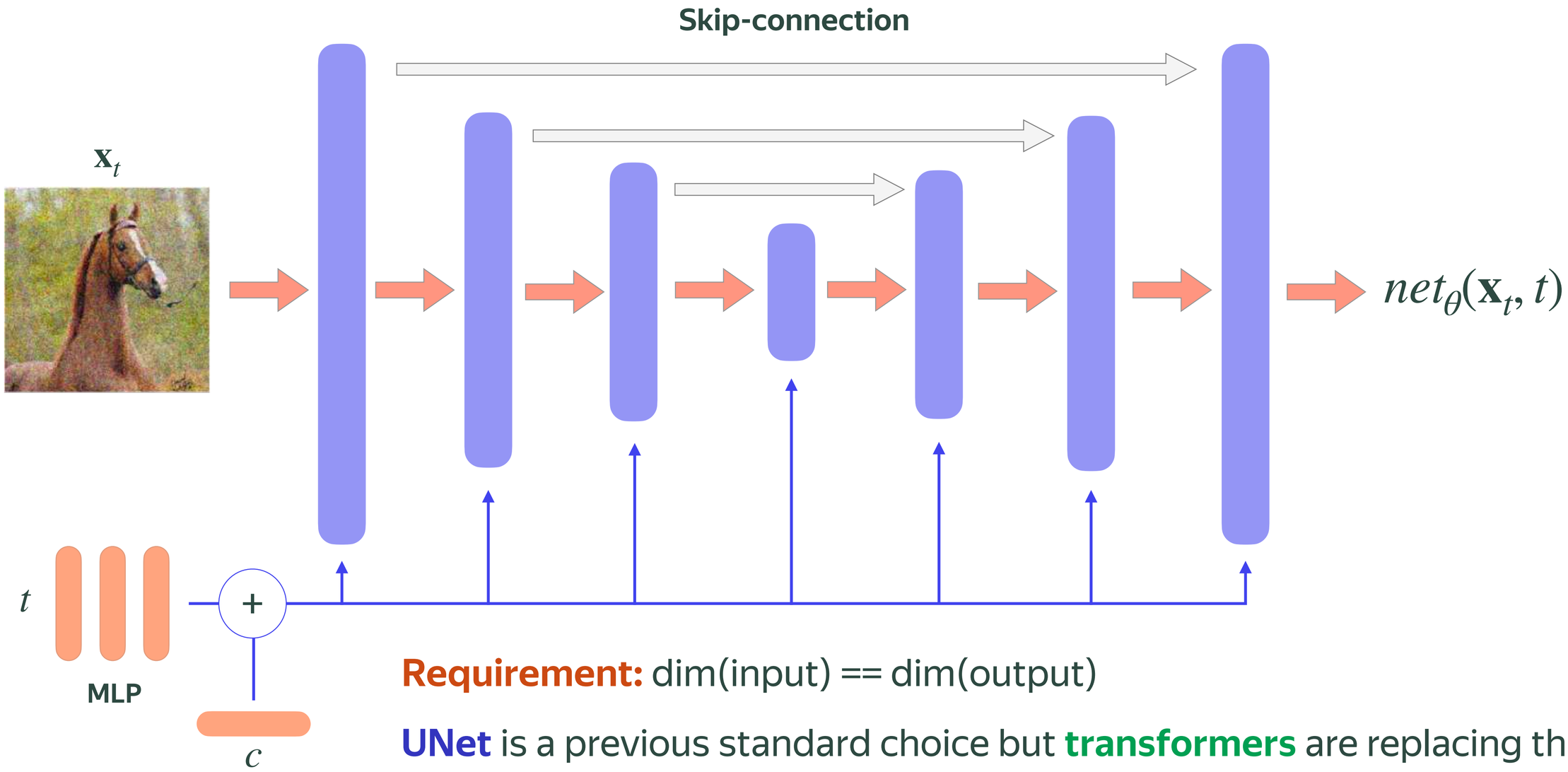
$$\mathbf{v}(\mathbf{x}_t) := \mathbb{E}\left[\mathbf{v}^{\text{cond}}(\mathbf{x}_t) \mid \mathbf{x}_t\right]$$ Intractable target

$$\mathbf{v}^{\text{cond}}(\mathbf{x}_t) := \dot{\mathbf{x}}_t = \dot{\alpha}_t \mathbf{x}_0 + \dot{\sigma}_t \boldsymbol{\epsilon} = \{\alpha_t = (1-t), \sigma_t = t\} = \boldsymbol{\epsilon} - \mathbf{x}_0$$

Conditional trick

$$\mathop{\mathbb{E}}_{t\sim p(t)} \mathop{\mathbb{E}}_{\mathbf{x}_t\sim p(\mathbf{x}_t)} \left[w(t) \cdot \|\mathbf{v}_\theta(\mathbf{x}_t, t) - \mathbf{v}(\mathbf{x}_t)\|_2^2\right] = \mathop{\mathbb{E}}_{t\sim p(t)} \mathop{\mathbb{E}}_{\mathbf{x}_t\sim p(\mathbf{x}_t)} \left[w(t) \cdot \|\mathbf{v}_\theta(\mathbf{x}_t, t) - \mathbf{v}^{cond}(\mathbf{x}_t)\|_2^2\right] + C$$

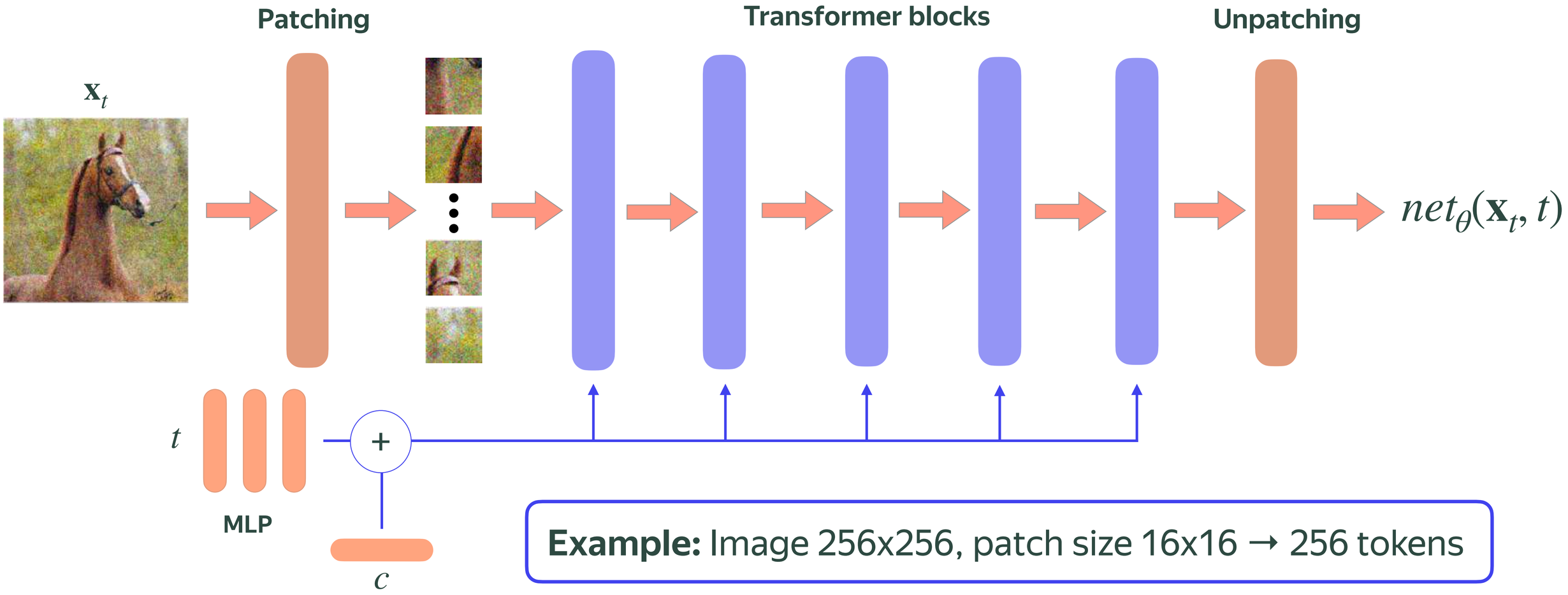
Diffusion UNet



Requirement: dim(input) == dim(output)

UNet is a previous standard choice but **transformers** are replacing them

Diffusion transformer



Example: Image 256x256, patch size 16x16 → 256 tokens

Sampling with PF-ODE

Score Matching

$$d\mathbf{x} = f(t)\mathbf{x} - \frac{1}{2}g^2(t)\, s_\theta(\mathbf{x}, t)dt$$

$$f(t) = \frac{d}{dt}\log \alpha_t = \frac{\dot{\alpha}_t}{\alpha_t}, \qquad g^2(t) = 2\sigma_t\dot{\sigma}_t - 2f(t)\sigma_t^2$$

$$d\mathbf{x} = \frac{\dot{\alpha}_t}{\alpha_t}\mathbf{x} - \sigma_t\left(\dot{\sigma}_t - \frac{\dot{\alpha}_t}{\alpha_t}\sigma_t\right) s_\theta(\mathbf{x}, t)dt = \{s_\theta(\mathbf{x}, t) = -\frac{\epsilon_\theta(\mathbf{x}, t)}{\sigma_t}\} = \boxed{\frac{\dot{\alpha}_t}{\alpha_t}\mathbf{x} + \left(\dot{\sigma}_t - \frac{\dot{\alpha}_t}{\alpha_t}\sigma_t\right)\epsilon_\theta(\mathbf{x}, t)dt}$$

Sampling with PF-ODE

Score Matching

Solve it! $d\mathbf{x} = \frac{\dot{\alpha}_t}{\alpha_t}\mathbf{x} + \left(\dot{\sigma}_t - \frac{\dot{\alpha}_t}{\alpha_t}\sigma_t\right)\epsilon_\theta(\mathbf{x}, t)dt$

Euler

DDIM

$$\mathbf{x}_{t-\Delta t} = \mathbf{x}_t - \Delta t\left(\underbrace{\frac{\dot{\alpha}_t}{\alpha_t}\mathbf{x}_t}_{\text{Linear part}} + \left(\dot{\sigma}_t - \frac{\dot{\alpha}_t}{\alpha_t}\sigma_t\right)\epsilon_\theta(\mathbf{x}_t, t)\right) \rightarrow \mathbf{x}_{t-\Delta t} = \alpha_{t-\Delta t}\left(\frac{\mathbf{x}_t - \sigma_t\epsilon_\theta(\mathbf{x}_t, t)}{\alpha_t}\right) + \sigma_{t-\Delta t}\,\epsilon_\theta(\mathbf{x}_t, t)$$

$$\dot{\alpha}_t = \frac{\alpha_t - \alpha_{t-\Delta t}}{\Delta t}, \quad \dot{\sigma}_t = \frac{\sigma_t - \sigma_{t-\Delta t}}{\Delta t}$$

DDIM == DPM-1 solver

Sampling | Flow matching

Solve ODE $d\mathbf{x} = \mathbf{v}_\theta(\mathbf{x}, t)dt$

Euler (1st-order)

$$\mathbf{x}_{t-\Delta t} = \mathbf{x}_t - \Delta t \cdot \mathbf{v}_\theta(\mathbf{x}_t, t)$$

Heun (2nd-order)

$$\tilde{\mathbf{x}}_{t-\Delta t} = \mathbf{x}_t - \Delta t \cdot \mathbf{v}_\theta(\mathbf{x}_t, t)$$

$$\mathbf{x}_{t-\Delta t} = \mathbf{x}_t - \frac{\Delta t}{2}\left(\mathbf{v}_\theta(\mathbf{x}_t, t) + \mathbf{v}_\theta(\tilde{\mathbf{x}}_{t-\Delta t}, t - \Delta t)\right)$$

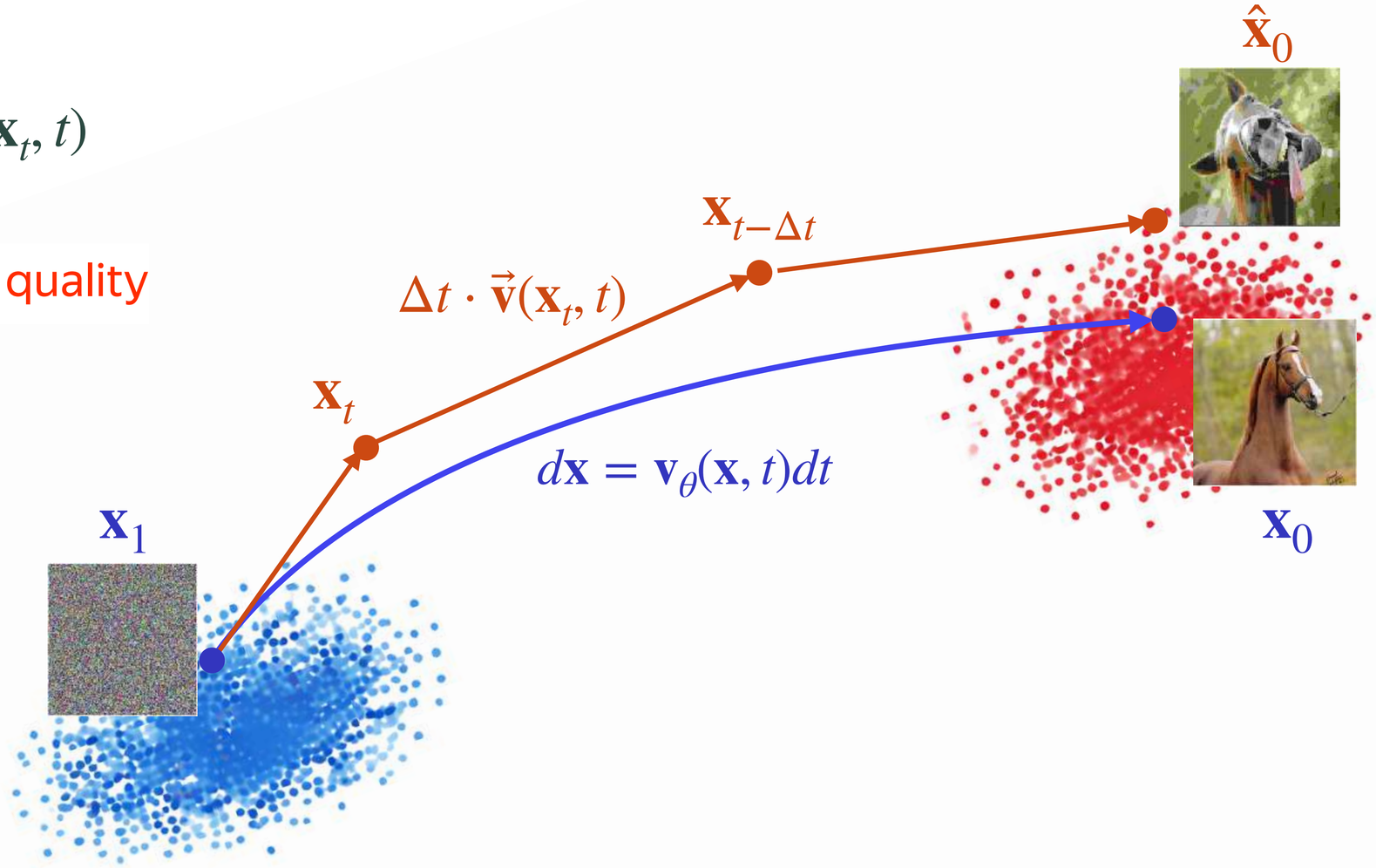
Euler / Heun == DPM-1 / DPM-2 solvers

* for flow matching

Speed-quality trade-off

Euler: $\mathbf{x}_{t-\Delta t} = \mathbf{x}_t - \Delta t \cdot \mathbf{v}_\theta(\mathbf{x}_t, t)$

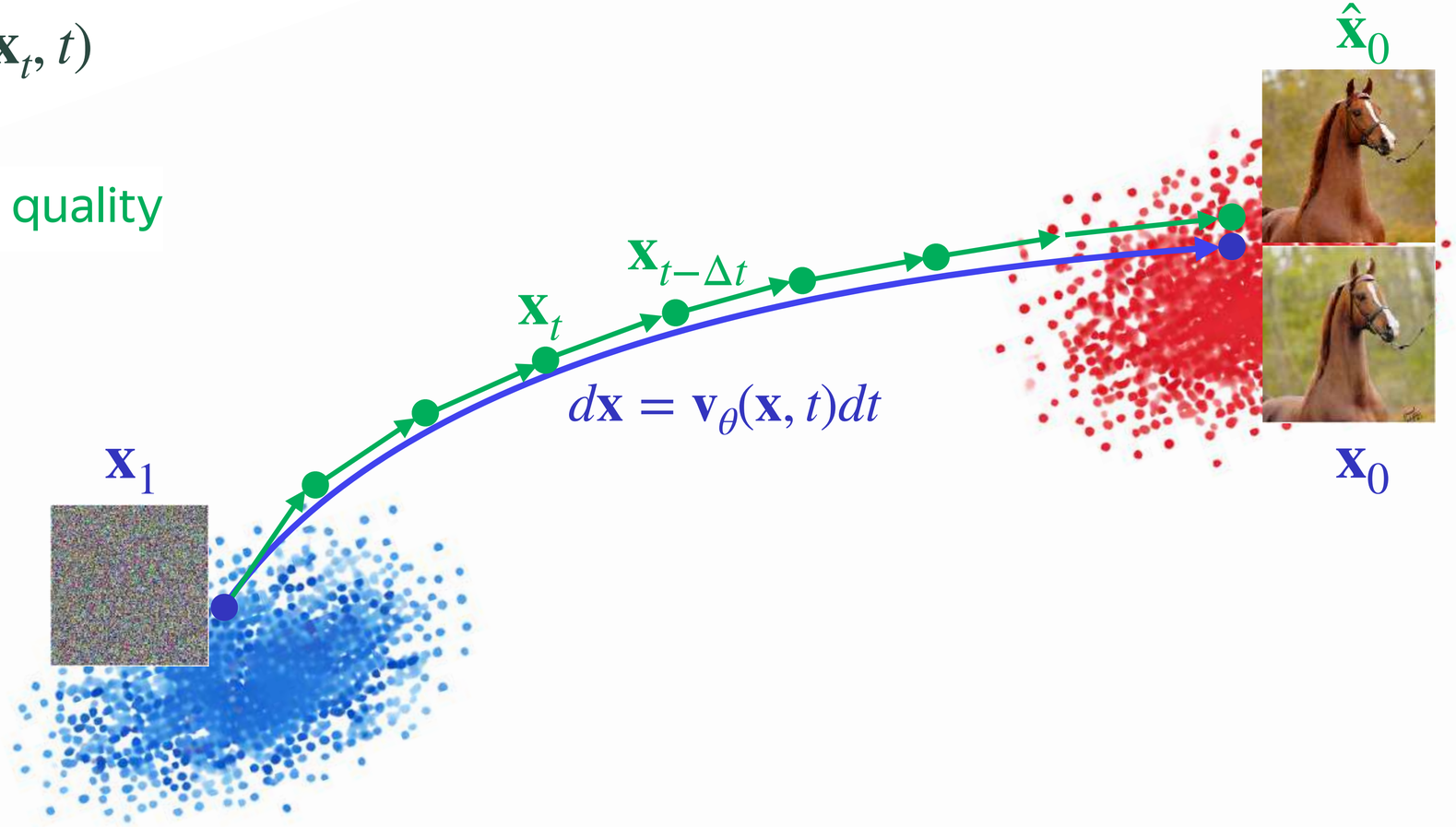
Large $\Delta t \rightarrow$ few steps & lower quality



Speed-quality trade-off

Euler: $\mathbf{x}_{t-\Delta t} = \mathbf{x}_t - \Delta t \cdot \mathbf{v}_\theta(\mathbf{x}_t, t)$

Small $\Delta t \rightarrow$ many steps & high quality



Practical note

2nd-order solvers can do better but
Euler is a default choice in practice

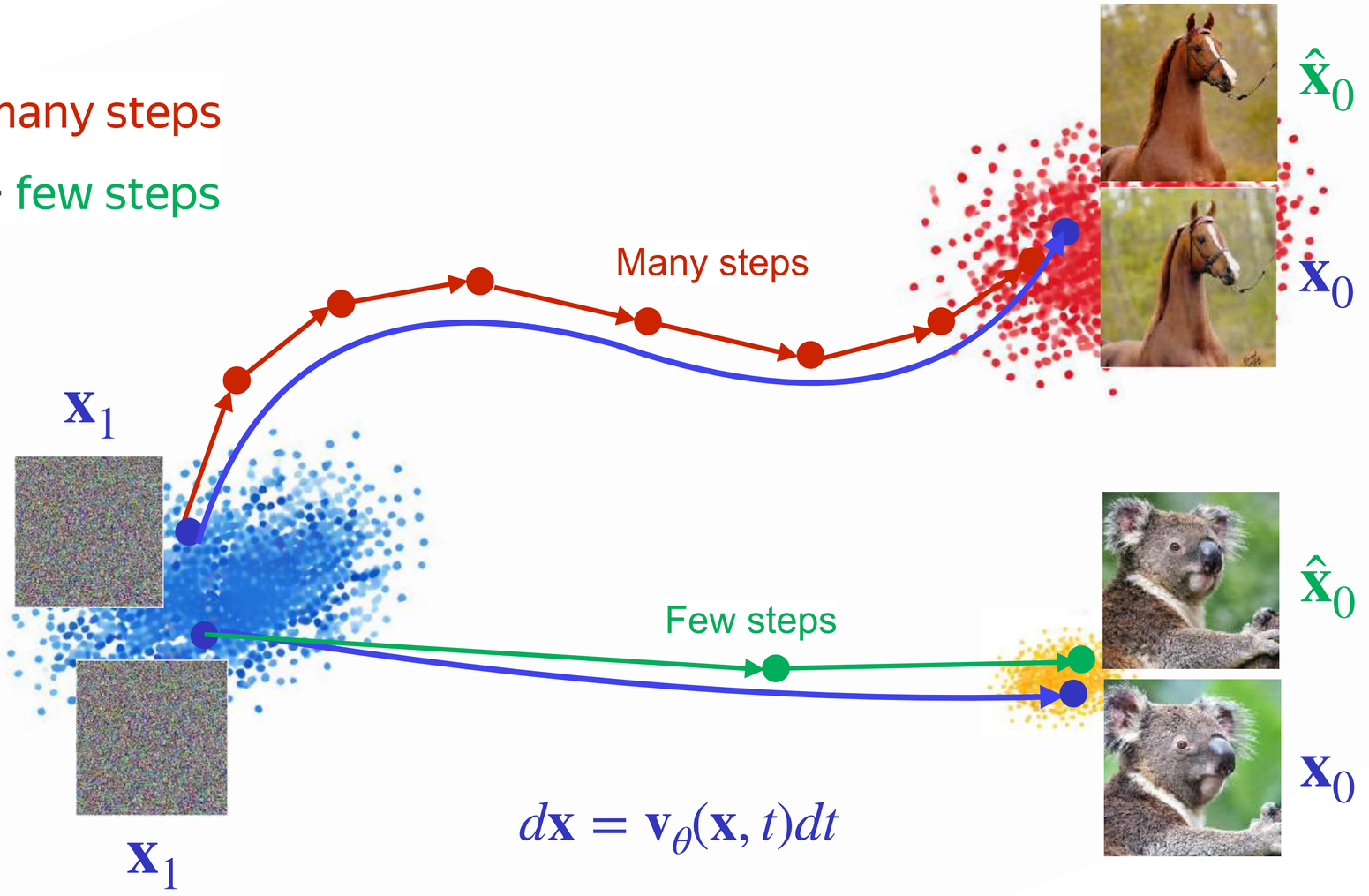
Effect of the trajectory curvature

Curvy trajectories —> small Δt —> many steps

Straight trajectories —> large Δt —> few steps

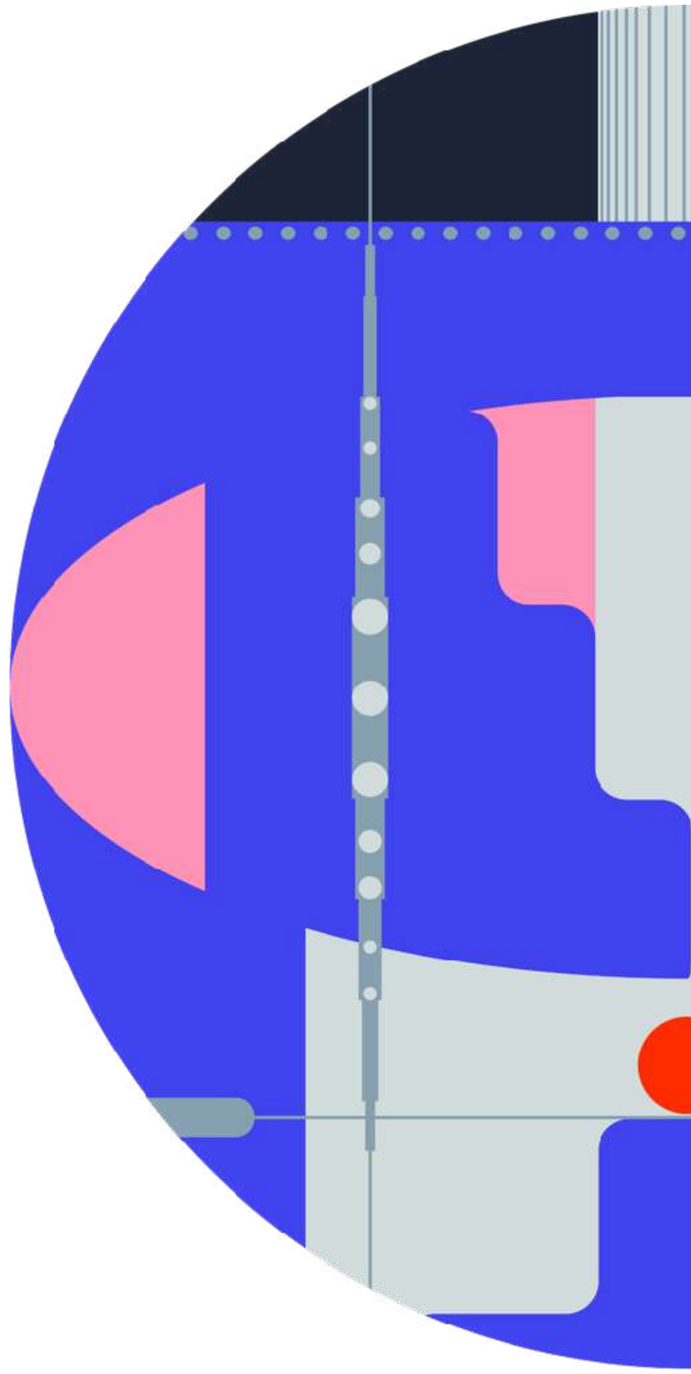
Practical note

The most effective way to get straighter trajectories — stronger condition, e.g., text or another image



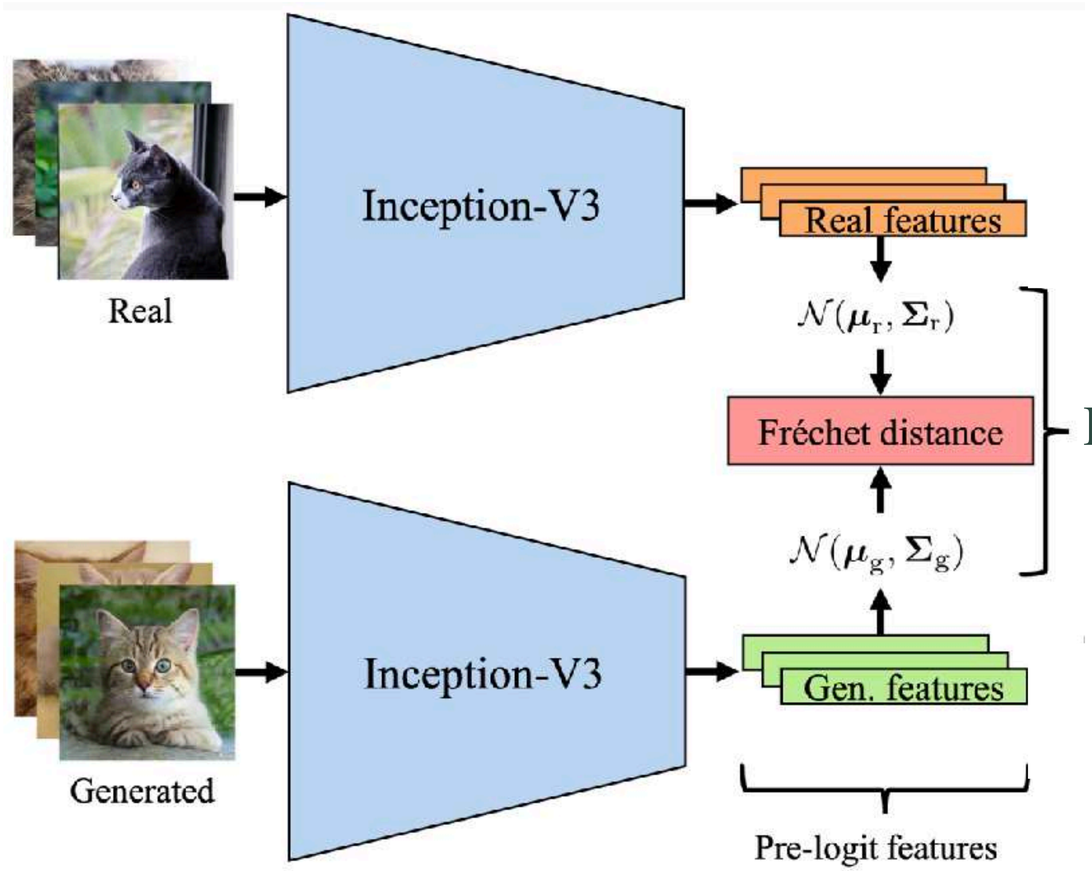
How to assess generative models?

*In academic scenarios



Fréchet Inception Distance (FID)

2-Wasserstein distance between real and fake **Gaussian** distributions in the **Inception V3** feature space



$$\mathrm{FID}(\mu_r, \Sigma_r, \mu_g, \Sigma_g) = \|\mu_r - \mu_g\|_2^2 + \mathrm{Tr}\left(\Sigma_r + \Sigma_g - 2(\Sigma_r\Sigma_g)^{1/2}\right)$$

Still widely used in papers

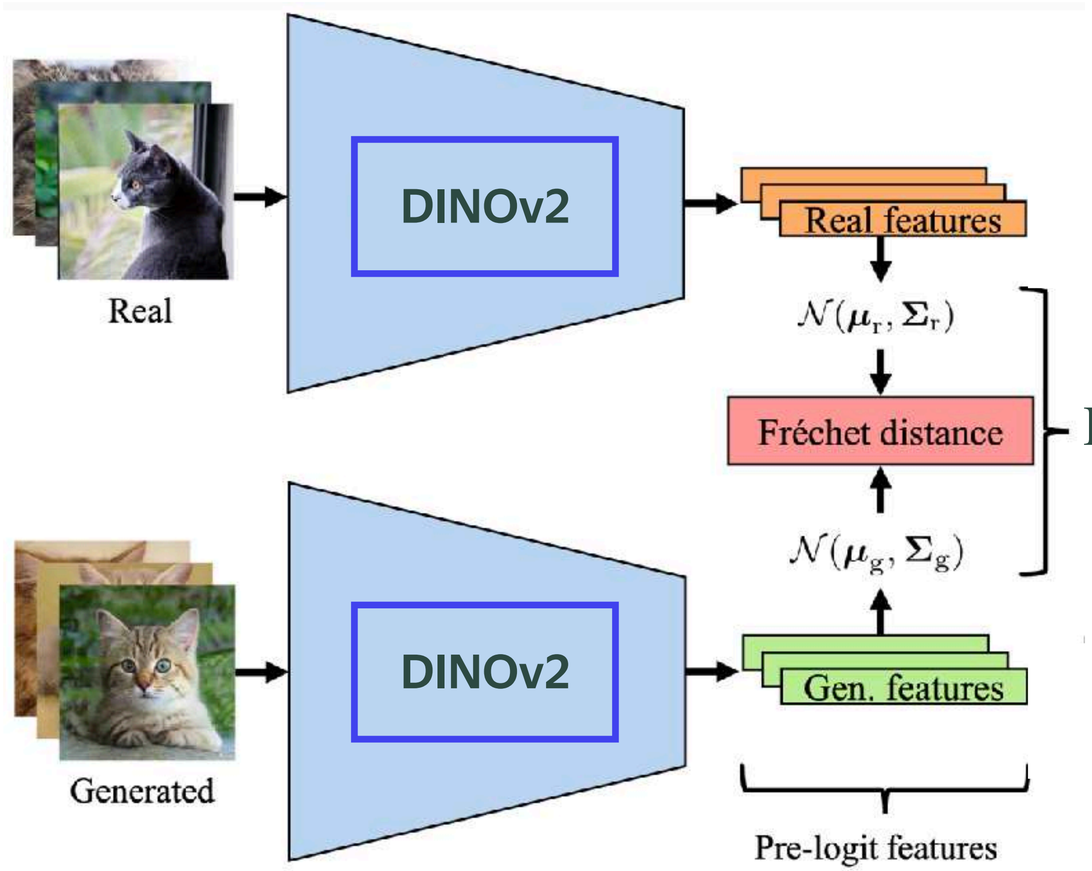
Works on deprecated Inception V3

Requires many samples: ~10K+

Gaussian assumption is weak today

Fréchet Distance DINOv2 (FDD)

2-Wasserstein distance between real and fake **Gaussian** distributions in the **DINOv2** feature space



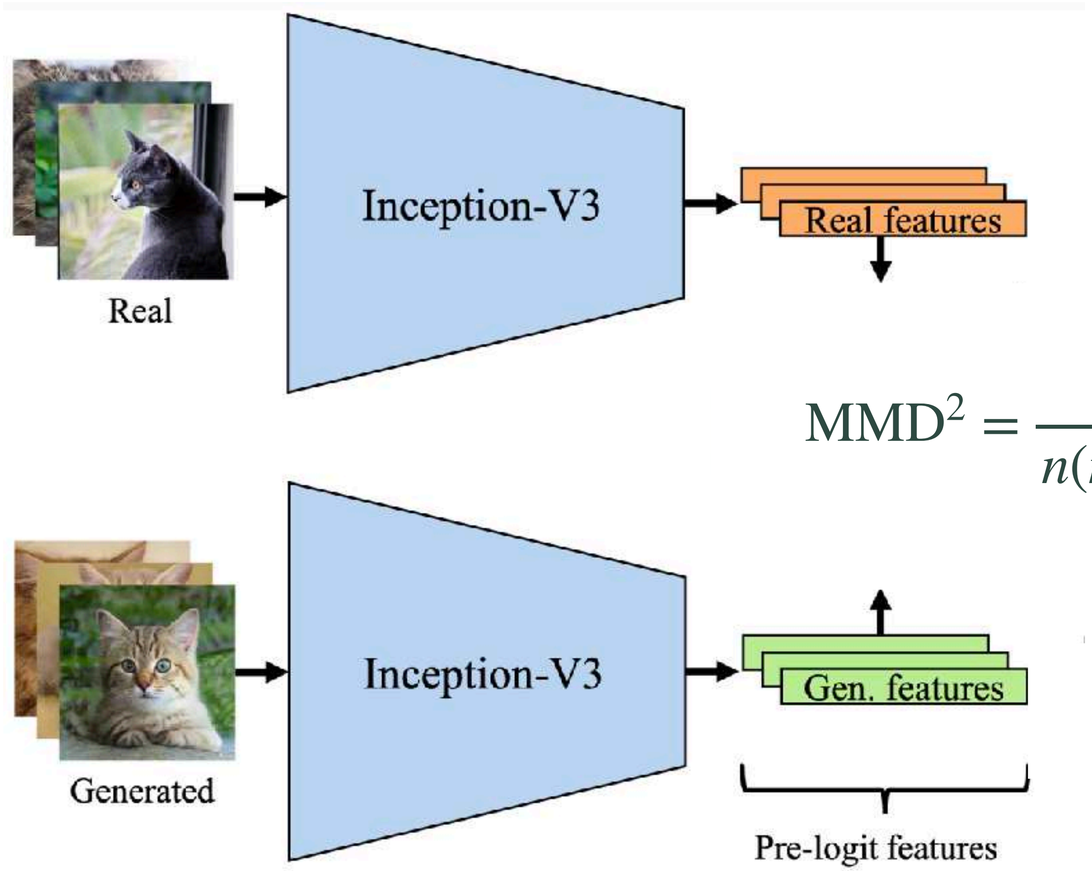
$$\mathrm{FDD}(\mu_r, \Sigma_r, \mu_g, \Sigma_g) = \|\mu_r - \mu_g\|_2^2 + \mathrm{Tr}\left(\Sigma_r + \Sigma_g - 2(\Sigma_r\Sigma_g)^{1/2}\right)$$

Requires many samples: ~10K+

Gaussian assumption is weak today

Kernel Inception Distance (KID)

Maximum Mean Discrepancy (MMD) between real and fake distributions in the **Inception V3** feature space



Idea: pairwise distances using $k(x, y)$

Polynomial kernel: $k(x, y) = \left(\frac{1}{d} x^\top y + 1\right)^3$

$$\mathrm{MMD}^2 = \frac{1}{n(n-1)} \sum_{i \neq j} k(x_i, x_j) + \frac{1}{m(m-1)} \sum_{i \neq j} k(y_i, y_j) - \frac{2}{nm} \sum_{i,j} k(x_i, y_j)$$

OK for much fewer samples: ~1K

No gaussian assumption anymore

Works on deprecated Inception V3

Takeaways

FID — legacy in papers. OK for academic datasets. **No way in s.o.t.a. text-to-image**

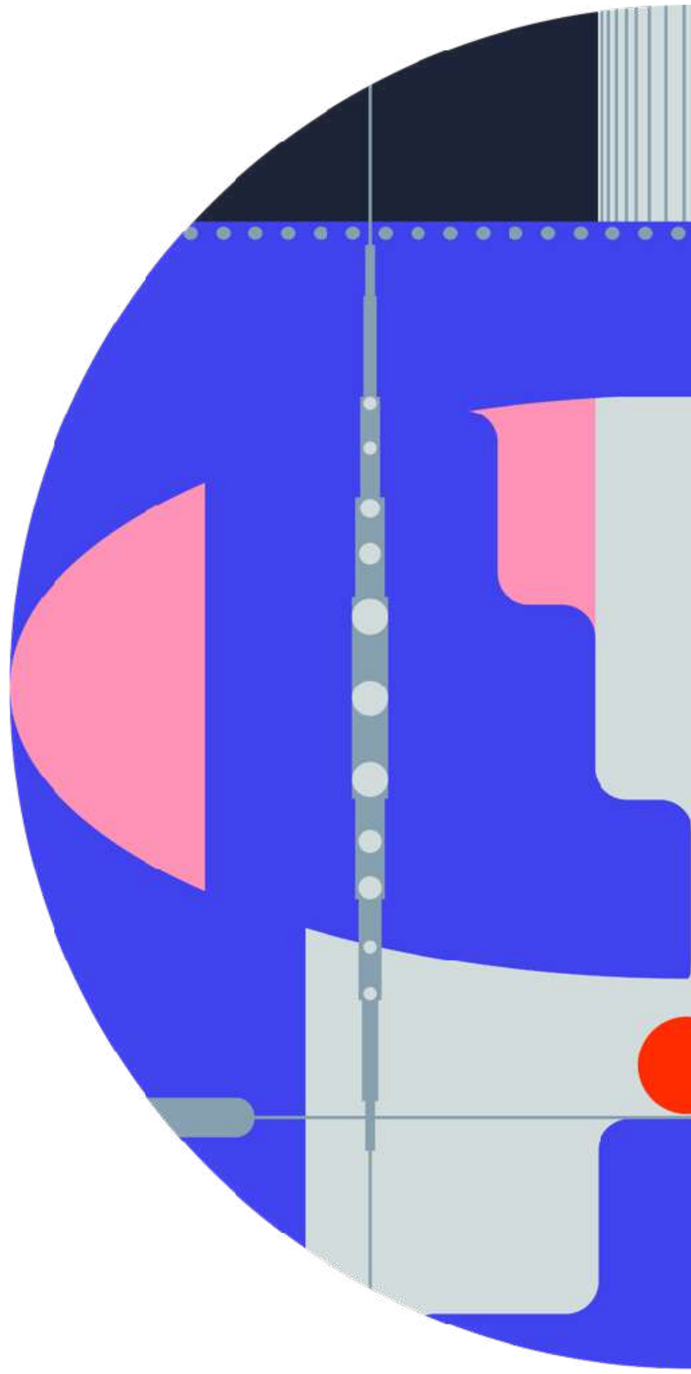
KID — used instead FID when few real/fake samples are available for evaluation

FDD — more recent and reliable FID variant but still not good enough for real-world cases

There are other metrics for academic cases like Precision and Recall

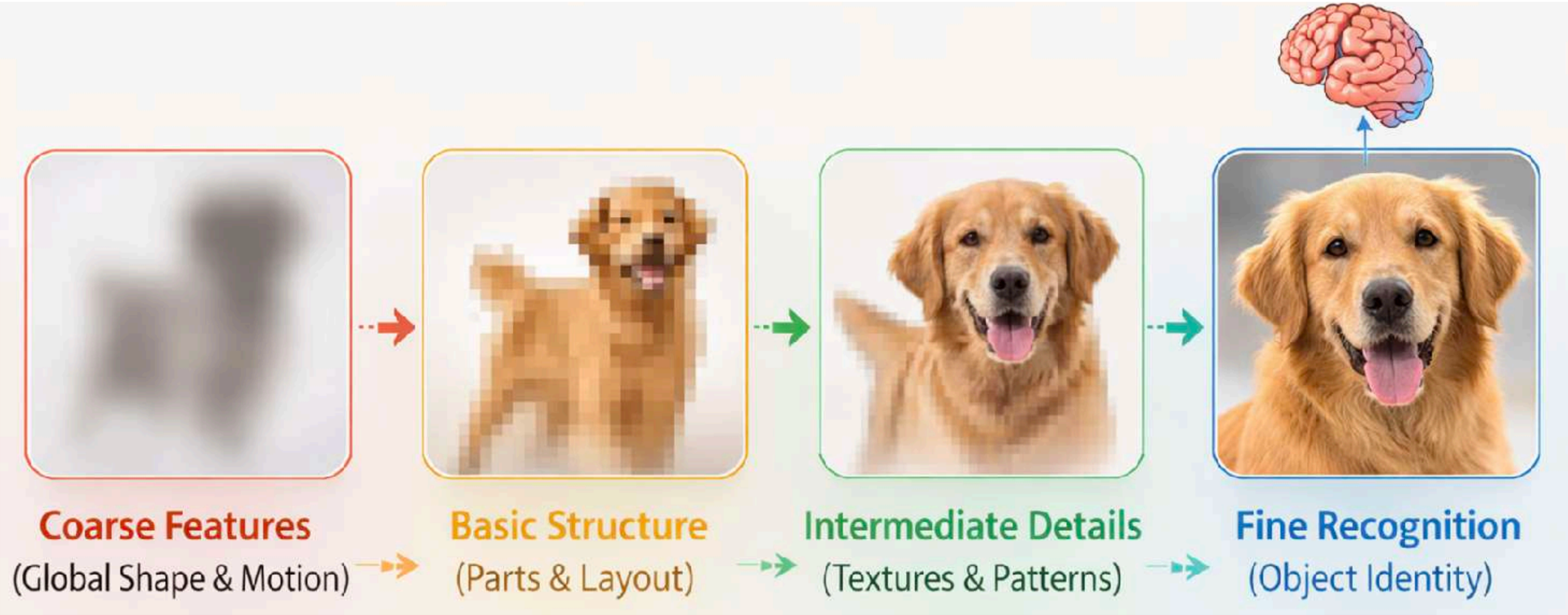
Text-to-image FID/KID variants, e.g., **CMMD,** seem reasonable but not quite popular

Why diffusion for visual generation?



How do people process image?

Human vision processes images hierarchically: from global structure to fine details



How do people generate images?

Pretty much similar...

FLUX.2

Diffusion also generates images in coarse-to-fine manner

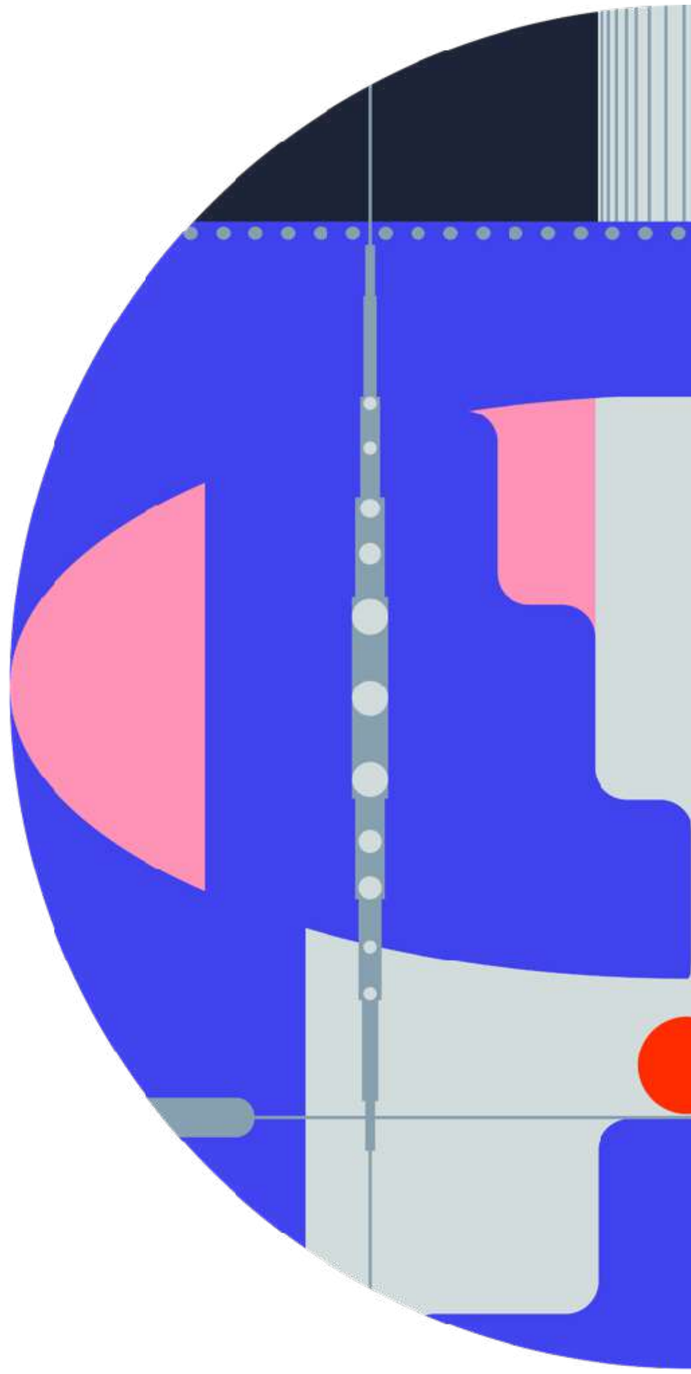
Layout and global shapes

Main object details

Fine-grained details

Textures, text, hair, grass

How to design your diffusion model?



How to design your diffusion model?

Design choices

1. Choice of $p(t)$ and $w(t)$
2. Network prediction $net_\theta(\mathbf{x}_t, t)$ for $\mathbf{v}_\theta$

Loss related

3. Diffusion space (pixel vs latent)
4. Representation alignment

Design related

Forward process: $\mathbf{x}_t = (1-t)\cdot\mathbf{x} + t\cdot\epsilon$

v-loss: $\mathop{\mathbb{E}}_{t\sim p(t)} \mathop{\mathbb{E}}_{\mathbf{x}_t\sim p(\mathbf{x}_t)} \left[w(t)\cdot\|v_\theta(\mathbf{x}_t, t) - \mathbf{v}\|_2^2\right]$

Loss-weighting vs Timestep distribution

Both $p(t)$ and $w(t)$ are equivalent for loss weighting

$$\mathop{\mathbb{E}}_{t\sim p(t)} \mathop{\mathbb{E}}_{\mathbf{x}_t\sim p(\mathbf{x}_t)} \left[\cancel{w(t)} \cdot \|v_\theta(\mathbf{x}_t, t) - \mathbf{v}\|_2^2 \right]$$

Always $w(t) \equiv 1$ and focus on $p(t)$

$p(t)$ gives more weight by sampling more frequently

$p(t)$ spends less compute on non-effective t

What is a good choice for $p(t)$?

Why intermediate timesteps matter

Intermediate timesteps are responsible for main content generation

Relatively easy

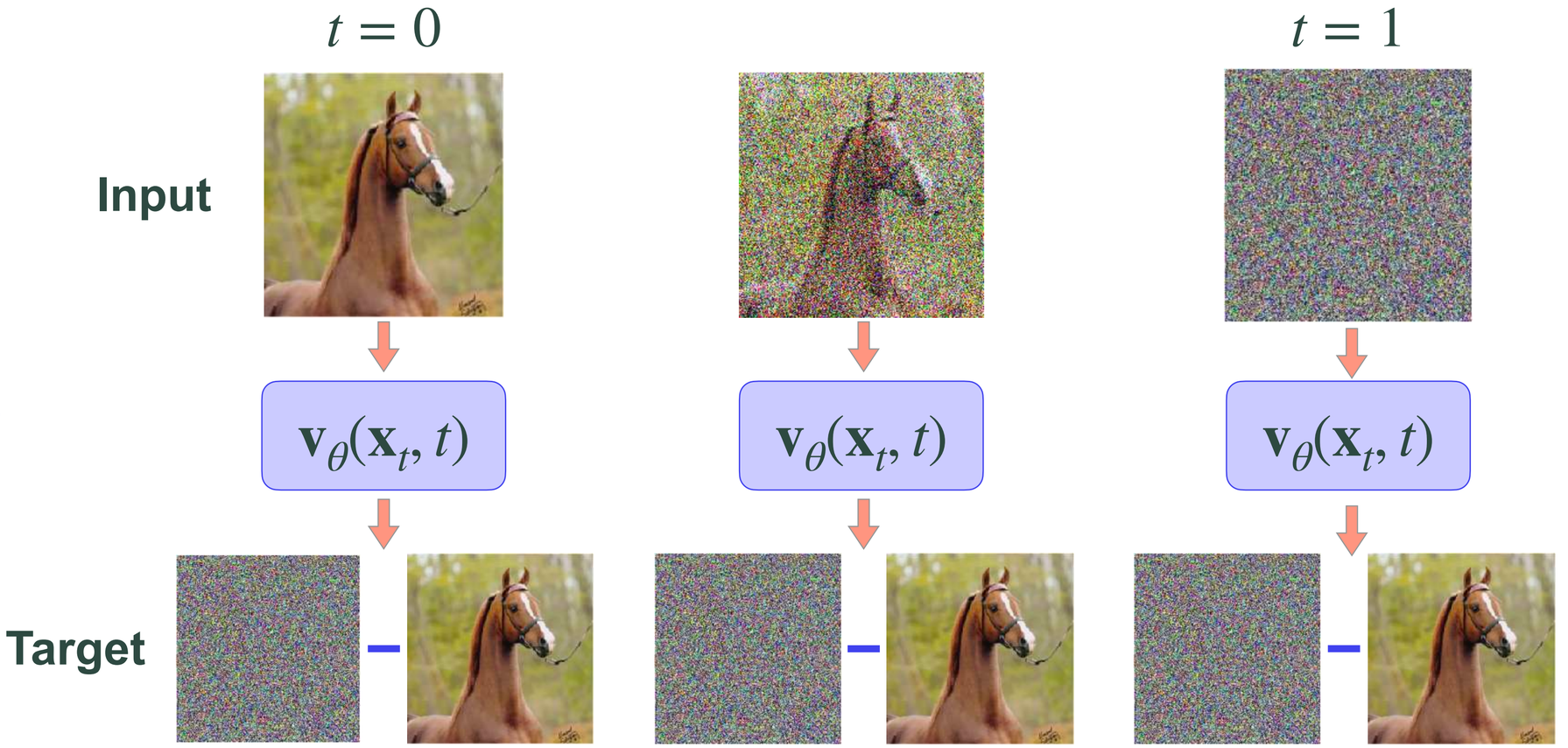
Difficult and important

Relatively easy

What do we make diffusion to predict?

$$\mathbb{E}\|\mathbf{v}_\theta(\mathbf{x}_t, t) - \mathbf{v}\|_2^2$$

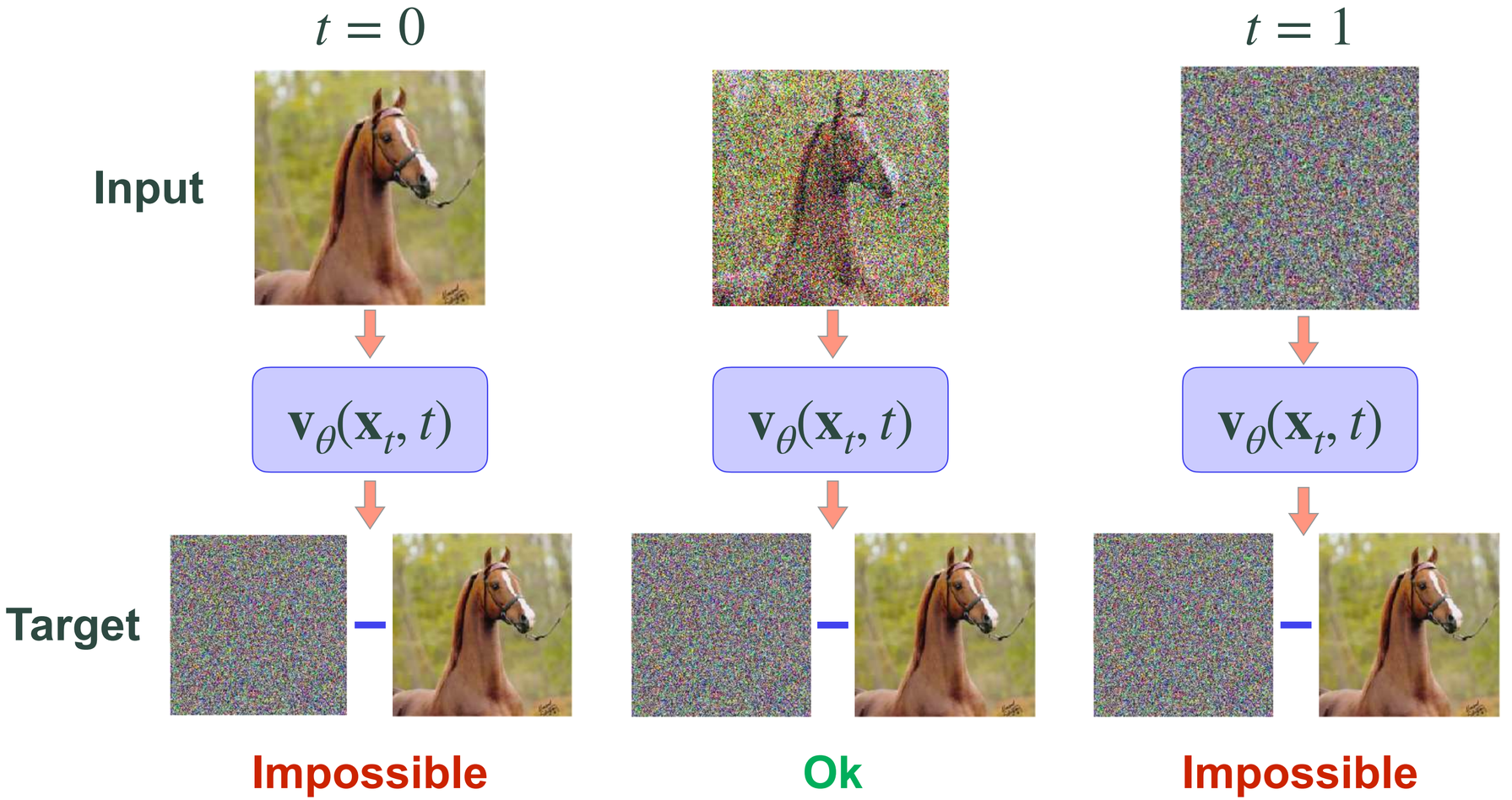
What are optimal predictions at $t = 0$ and $t = 1$?



What do we make diffusion to predict?

$$\mathbb{E}\|\mathbf{v}_\theta(\mathbf{x}_t, t) - \mathbf{v}\|_2^2$$

Intermediate timesteps are more reasonable to learn



Effective timestep distribution

Let’s make $p(t)$ focus on intermediate timesteps

Logit-normal distribution

$$p_{\ln}(t;\mu,\sigma) = \frac{1}{\sigma\sqrt{2\pi}\,t(1-t)} \exp\left(-\frac{(\operatorname{logit}(t)-\mu)^2}{2\sigma^2}\right)$$

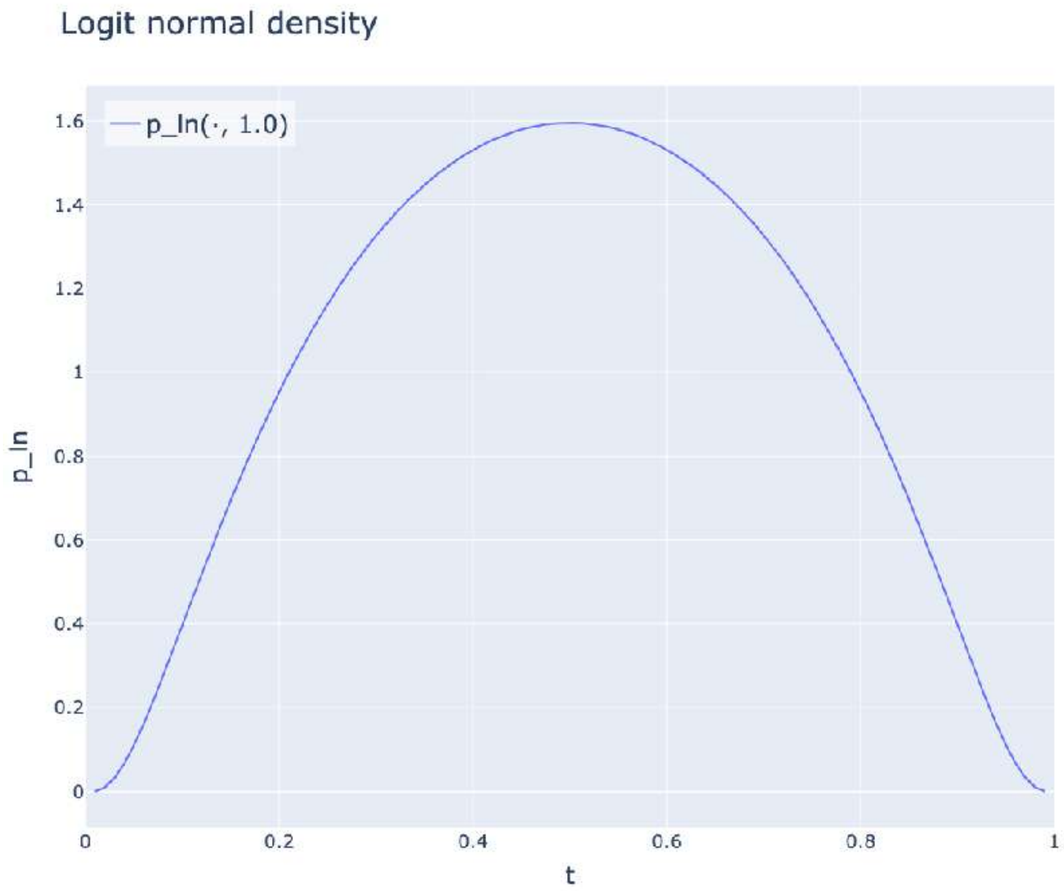
$$\operatorname{logit}(t) = \log\left(\frac{t}{1-t}\right) \qquad \mu = \log\alpha, \quad \sigma = 1$$

Effective timestep distribution

Let’s make $p(t)$ focus on intermediate timesteps

Logit-normal distribution ***in practice***

$$p_{\ln}(t;\alpha) = \frac{1}{\sqrt{2\pi}\, t(1-t)} \exp\left(-\frac{(\operatorname{logit}(t) - \log \alpha)^2}{2} \right)$$



FLUX.2: Analyzing and Enhancing the Latent Space of FLUX - Representation Comparison, 2025

Effective timestep distribution

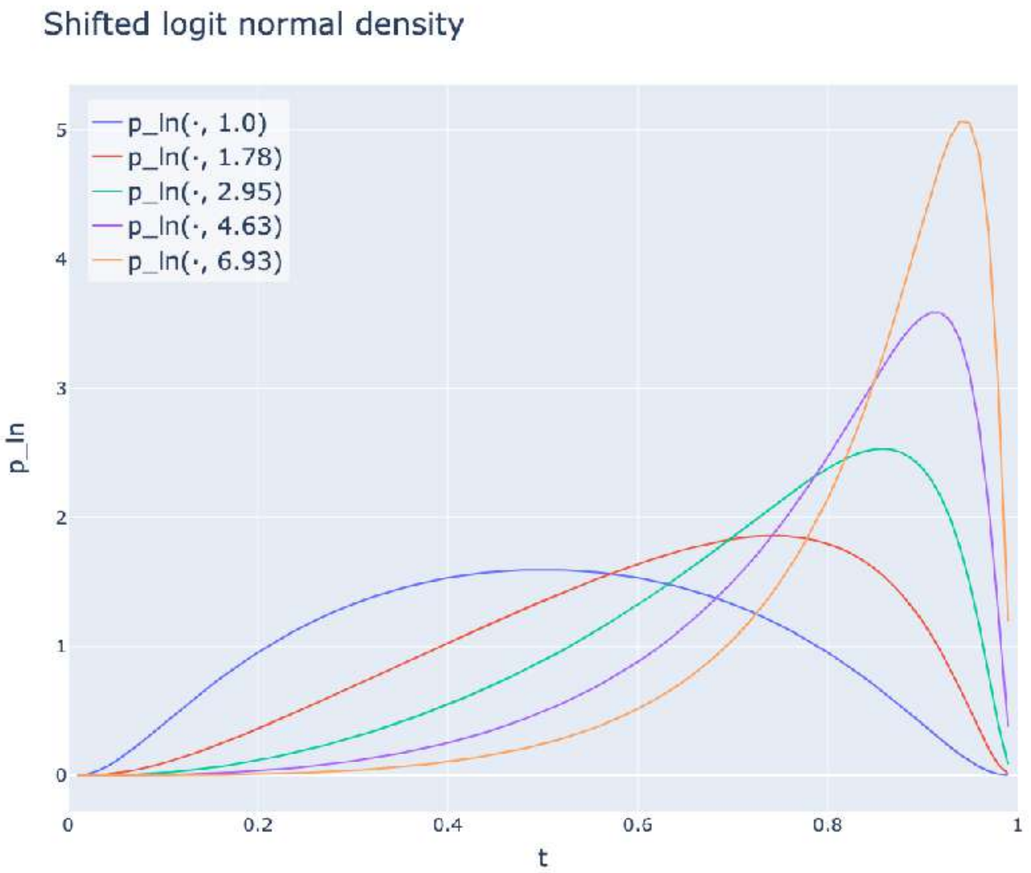
Let's make $p(t)$ focus on intermediate timesteps

Logit-normal distribution ***in practice***

$$p_{\ln}(t;\alpha) = \frac{1}{\sqrt{2\pi}\, t(1-t)} \exp\left(-\frac{(\operatorname{logit}(t) - \log \alpha)^2}{2} \right)$$

Often, it is beneficial to **shift** it to high noise levels

$$\alpha > 1$$



FLUX.2: Analyzing and Enhancing the Latent Space of FLUX - Representation Comparison, 2025

Why shifting $p(t)$?

Goal: understand what happens with $\mathbf{x}_t = (1-t)\mathbf{x}_0 + t\epsilon$ at **different image resolutions**

Why shifting $p(t)$?

Proxy for a real image $\mathbf{x}_0$

Let's consider a dummy random variable $Y(t) = (1-t)c + t\epsilon$, where $c \in \mathbb{R}, \epsilon \sim \mathcal{N}(0,1)$

$\mathbb{E}\left[Y(t)\right] = \mathbb{E}\left[(1-t)c + t\epsilon\right] = (1-t)c \rightarrow$ given N observations of $Y(t)$: $\hat{c} = \frac{1}{1-t}\frac{1}{N}\sum_i^N y_i$

$Var\left[Y(t)\right] = Var[(1-t)c + t\epsilon] = t^2 \rightarrow Var\left[\hat{c}\right] = \frac{1}{(1-t)^2}\frac{1}{N^2}\sum_i^N Var\left[y_i\right] = \frac{t^2}{(1-t)^2}\frac{1}{N}$

Why shifting $p(t)$?

$$Var\left[Y(t)\right] = \frac{t^2}{(1-t)^2}\frac{1}{N} \rightarrow \sigma(t,N) = \frac{t}{1-t}\sqrt{\frac{1}{N}} \rightarrow$$ More samples N reduce uncertainty as $\sqrt{\frac{1}{N}}$

Same noise level t destroys data $\mathbf{x}_0$ less at higher resolution

$t = 0.25$

128x128

256x256

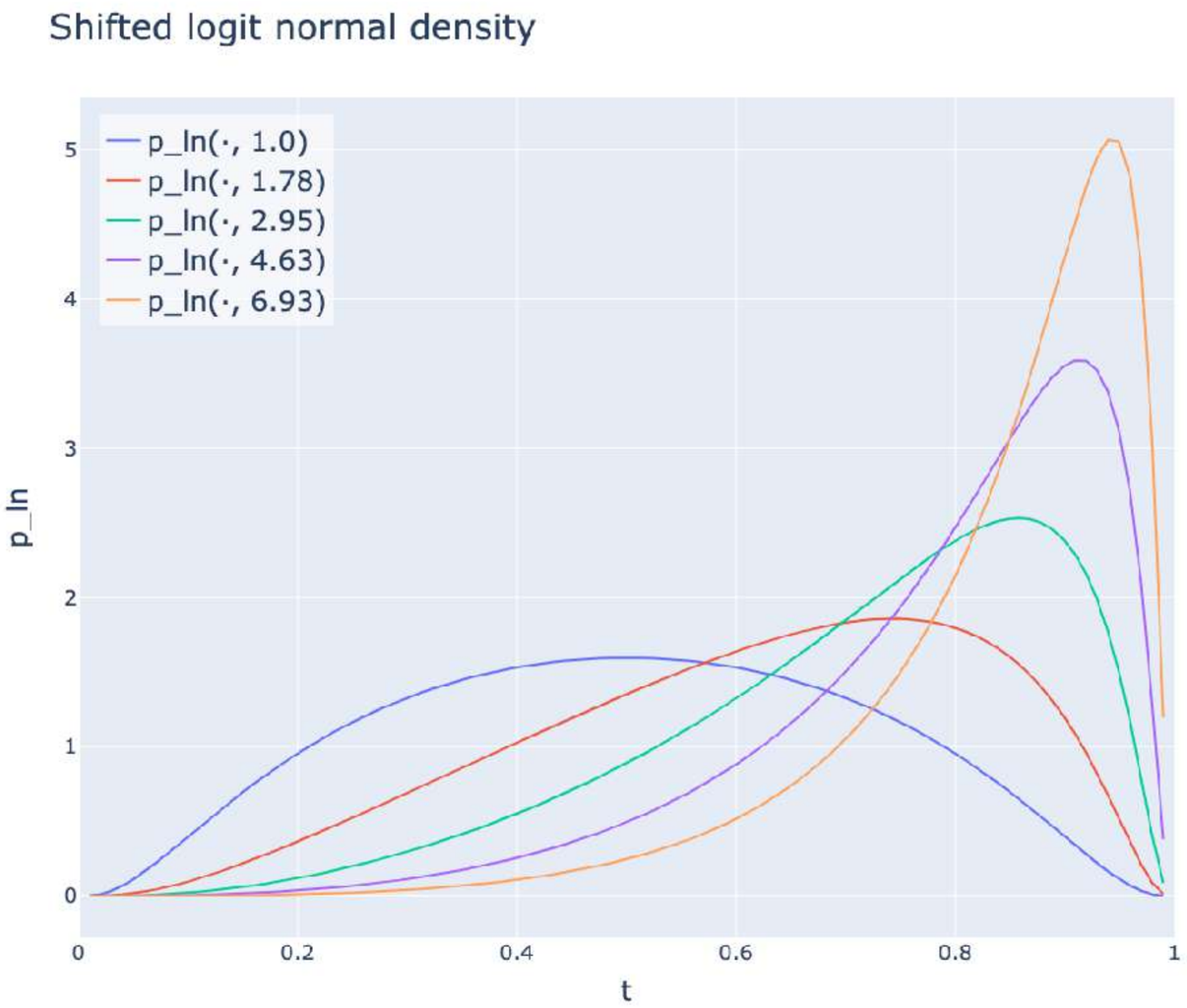
512x512

1024x1024

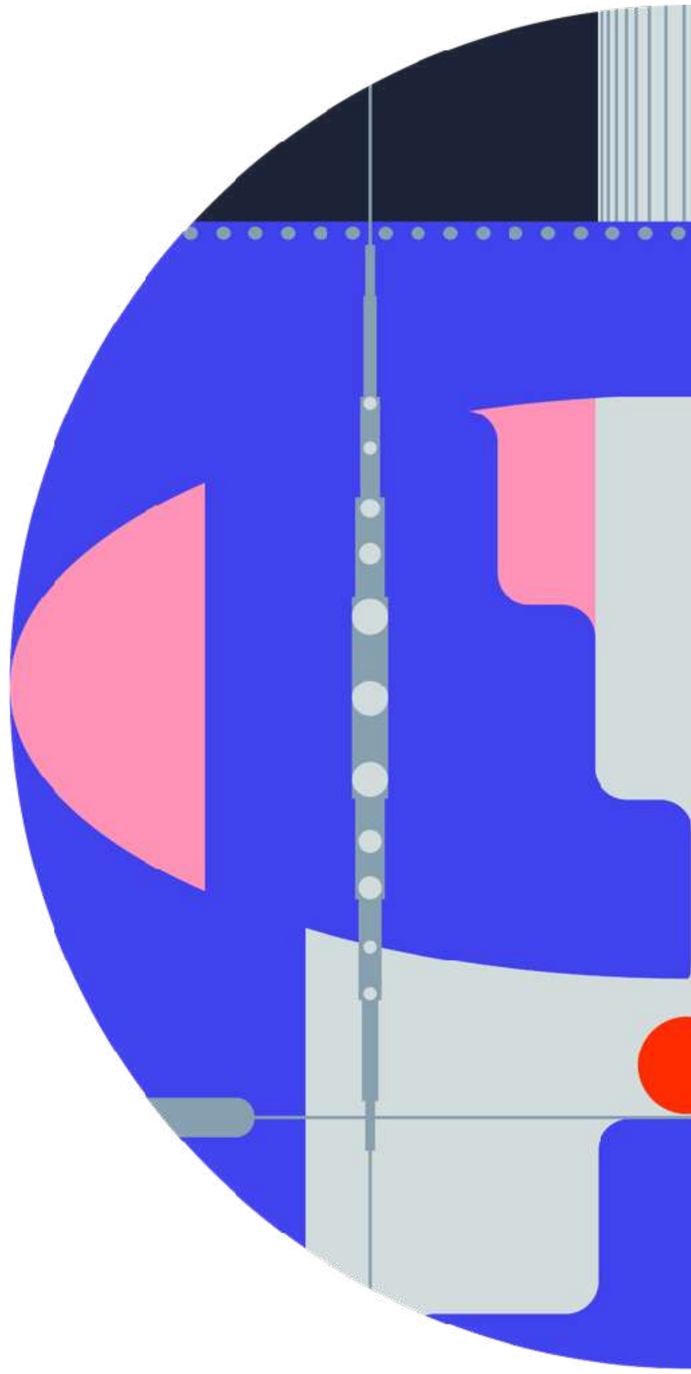
Takeaway

$$p_{\ln}(t;\alpha) = \frac{1}{\sqrt{2\pi}\, t(1-t)} \exp\left(-\frac{(\operatorname{logit}(t) - \log\alpha)^2}{2} \right)$$

Higher resolution $\rightarrow$ larger shift α



Diffusion spaces



Network prediction in pixel space

Noise in ϵ, $\mathbf{v}$ make target full dimensional

↓

Difficult for $net_\theta(\mathbf{x}_t, t)$ to learn it

$\mathbf{x}_0$ lies on lower dimensional subspace

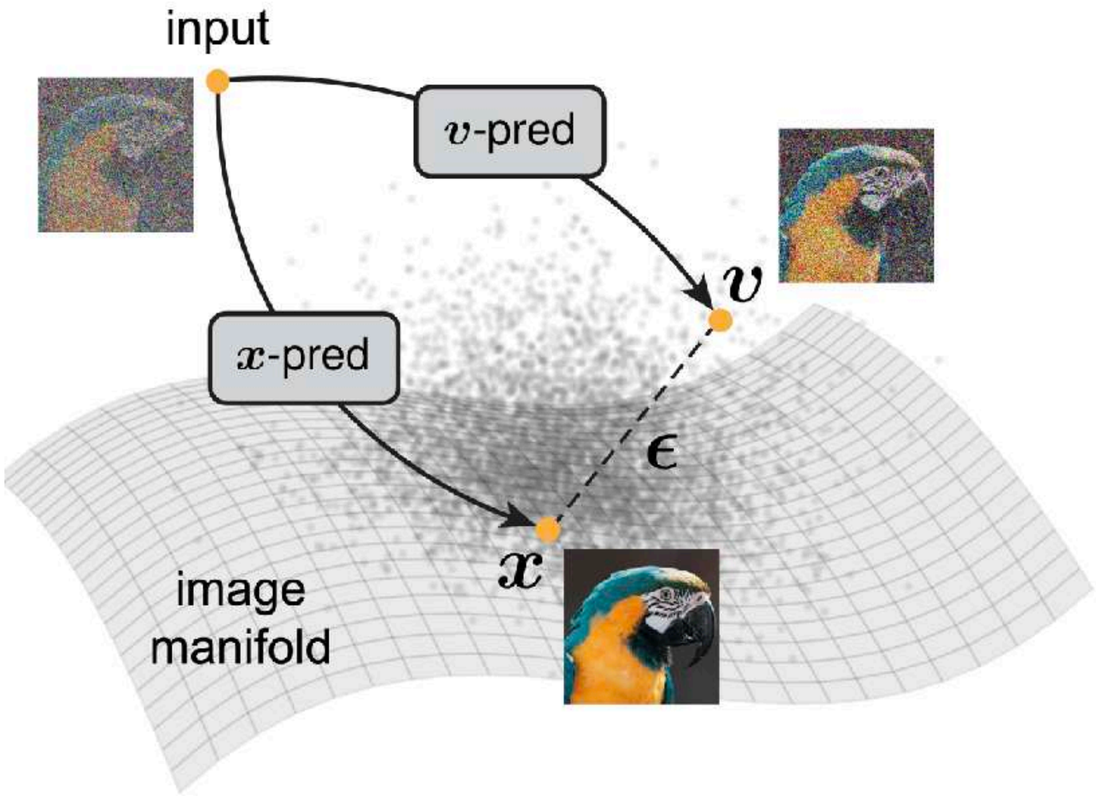
↓

Easier for $net_\theta(\mathbf{x}_t, t)$ to learn it

Pixel-space diffusion → $\mathbf{x}_0^\theta := net_\theta(\mathbf{x}_t, t)$

Manifold Assumption

Natural images lie on a low-dimensional manifold in the high-resolution pixel space



T. Li, K. He, Back to Basics: Let Denoising Generative Models Denoise, 2025

Is it really critical?

$\mathbf{x}_0$-prediction is **must-have** at high resolution

Dataset	$\mathbf{x}_0$-pred	ϵ-pred	$\mathbf{v}$-pred
ImageNet 256×256	8.62	372.38	96.53
ImageNet 64×64	3.55	3.63	3.46

FID results for different pred types

Problem

Even $\mathbf{x}_0$ is relatively high dimensional

Pixel space models are not s.o.t.a. **yet**

What is the problem with pixel space?

Natural images contain much imperceptible noise and high-frequency details

↓

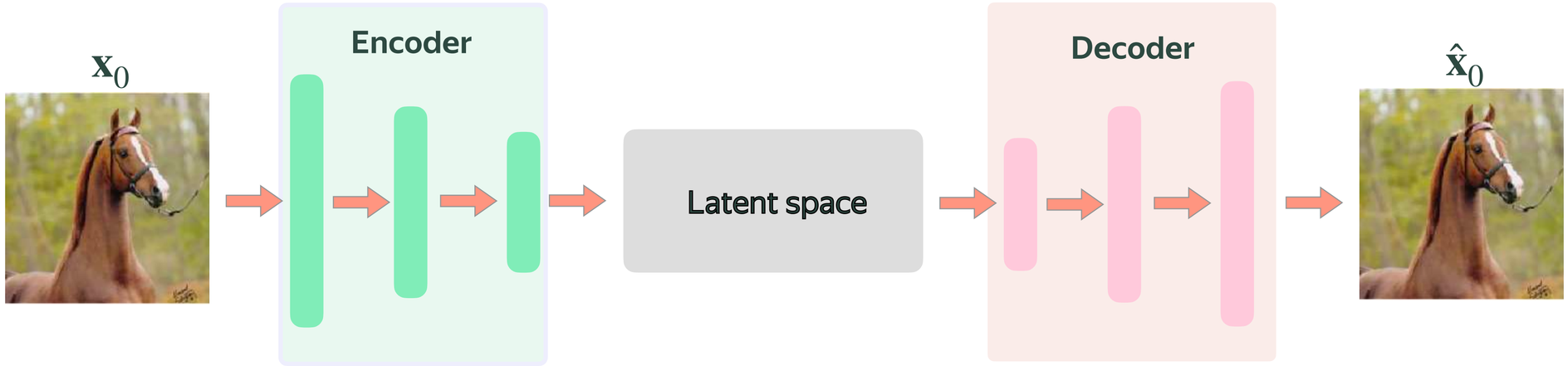
Natural images remain relatively high dimensional

↓

Diffusion models spend much capacity to fit this

Latent diffusion models

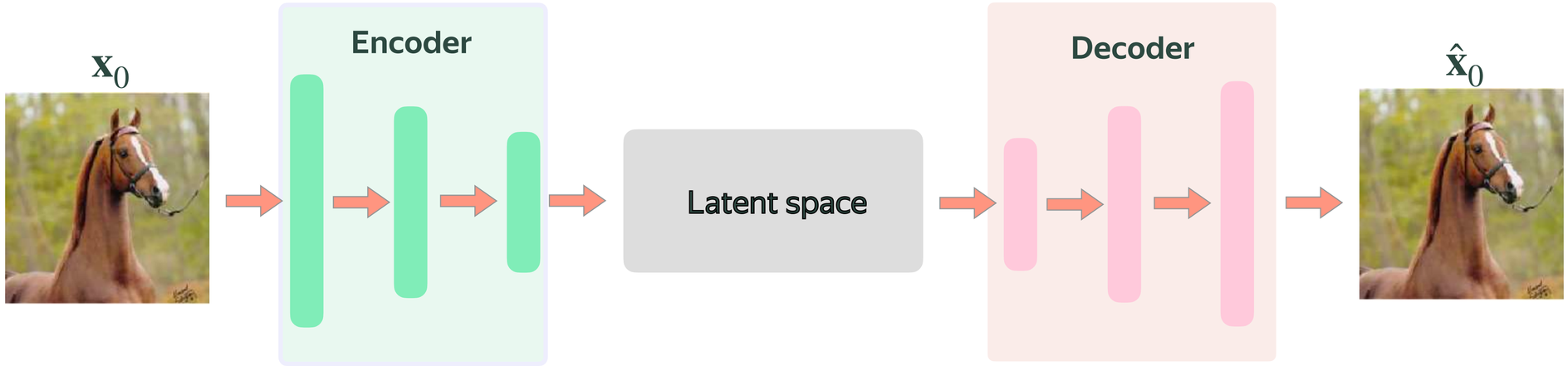
Use VAE to map images into lower dimensional latent space



R. Rombach, High-Resolution Image Synthesis with Latent Diffusion Models, 2022

Latent diffusion models

The V in VAE is vestigial: in fact, it just AE with slight regularization



Reconstruction AE loss: $L_{AE} = L_{rec} + L_{lpips} + L_{GAN}$

S. Dieleman, Generative modeling in latent space, 2025

FLUX.2 VAE latent space

Image

VAE latent channels

Latents are more like
advanced pixels

Quality-Learnability-Compression trade-off

What do we want from a VAE for diffusion?

Quality (Top priority)

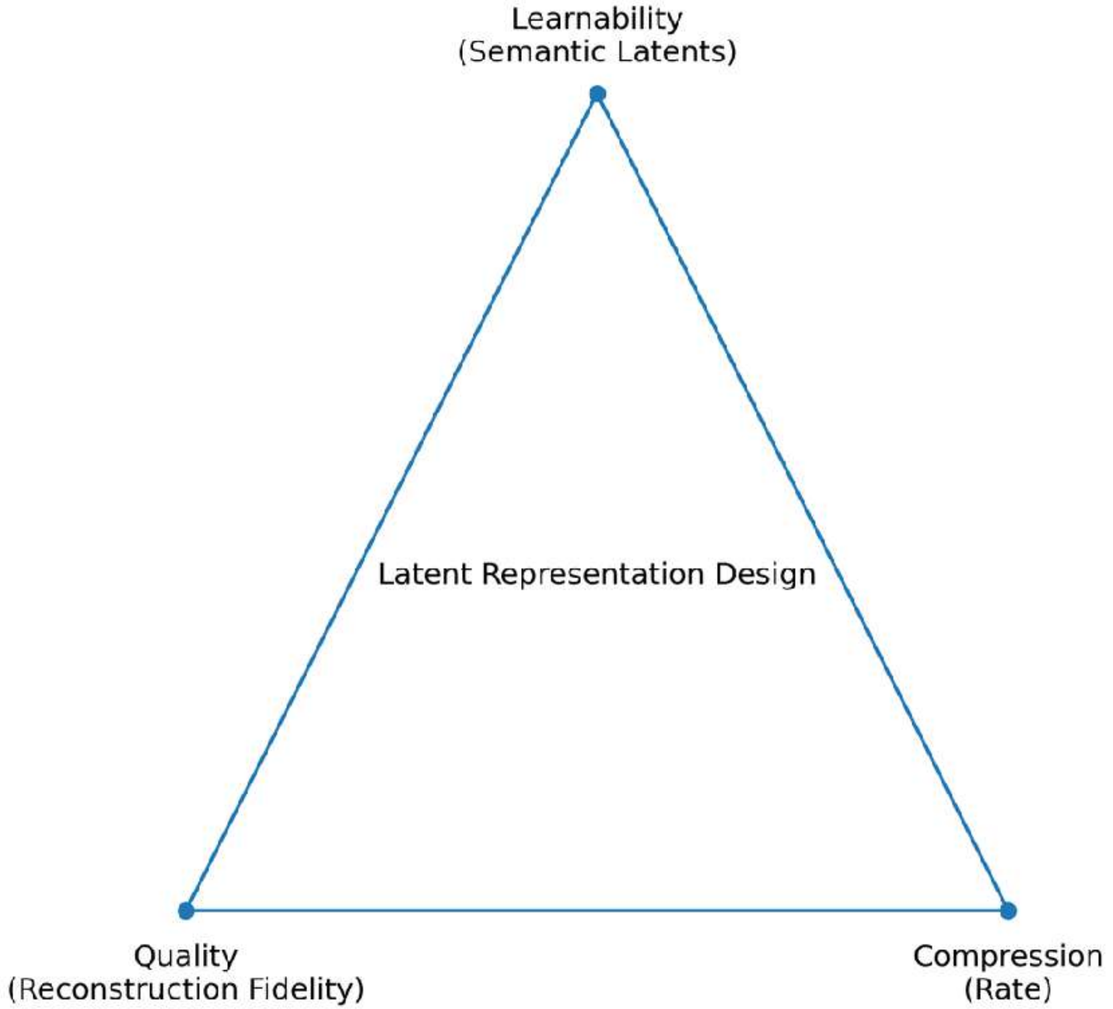
Nearly perfect image reconstruction

Learnability

Semantically structured latents make diffusion easier to learn

Compression

Lower-dimensional latent space for computational efficiency



How to enforce learnability?

The same image can be encoded in many different ways → find the more convenient one for diffusion

Apply latent space regularization

Examples

1. VAE-like prior: $D_{KL}(p_\theta(z)\|\mathcal{N}(0,I))$ — make latents closer to standard Gaussian

2. Scale equivariance: similar latent at different image resolutions

3. Representation losses: bias the encoder toward more **semantic structure**

Takeaways

Pixel-space models are promising and actively developed at the moment

$\mathbf{x}_0$-prediction is critical in pixel space. In latent space, network prediction type is less important

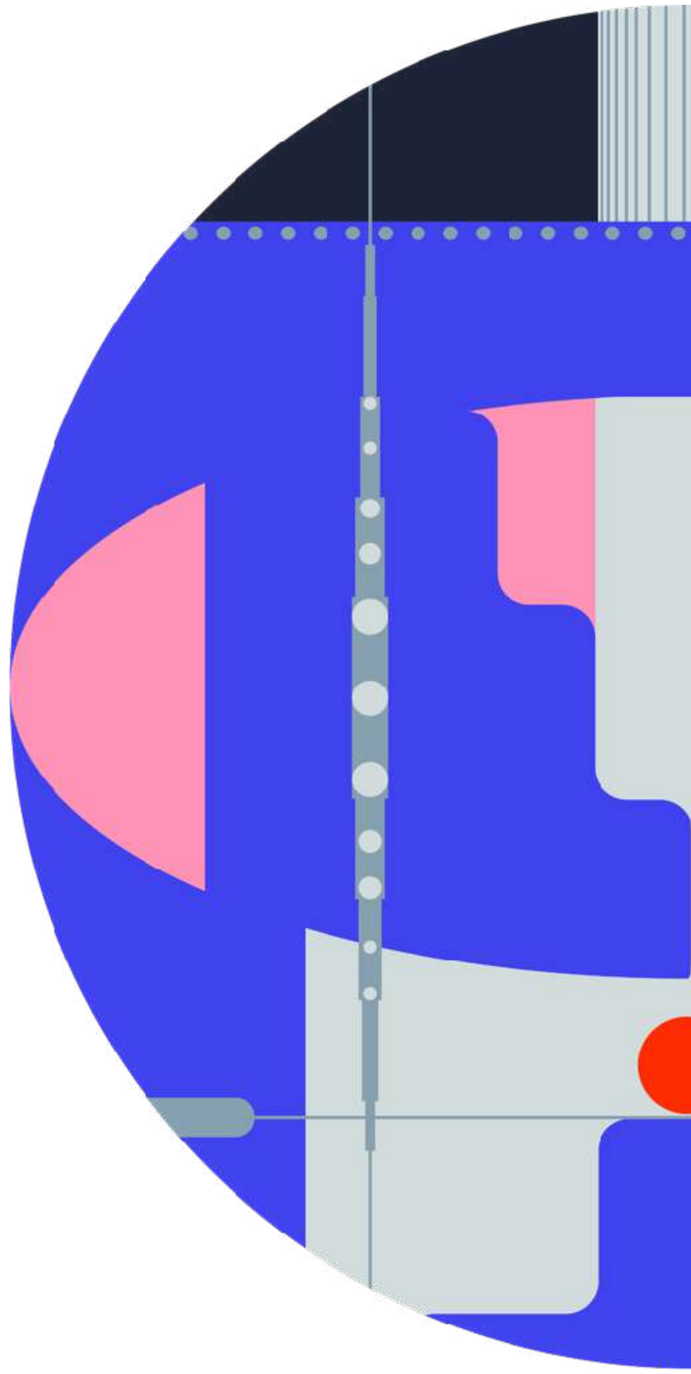
Latent diffusion models are a default choice in production-grade models

AE takes care of imperceptible noise and high-frequency details → low-dim latent space → higher learnability

AE latent space can be regularized to pack images in a more convenient way for diffusion

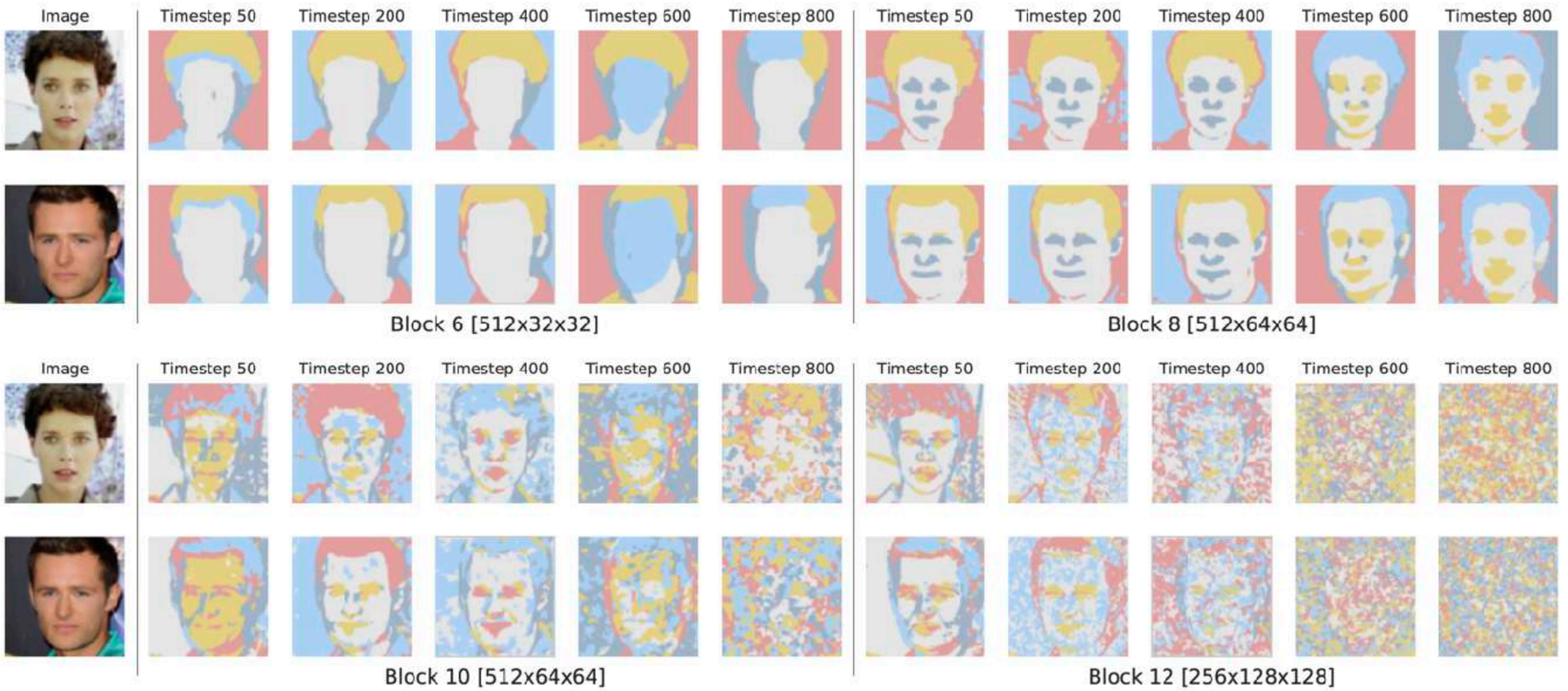
Designing AE with nearly perfect reconstruction and high learnability is a high-priority problem today

Diffusion models as representation learners



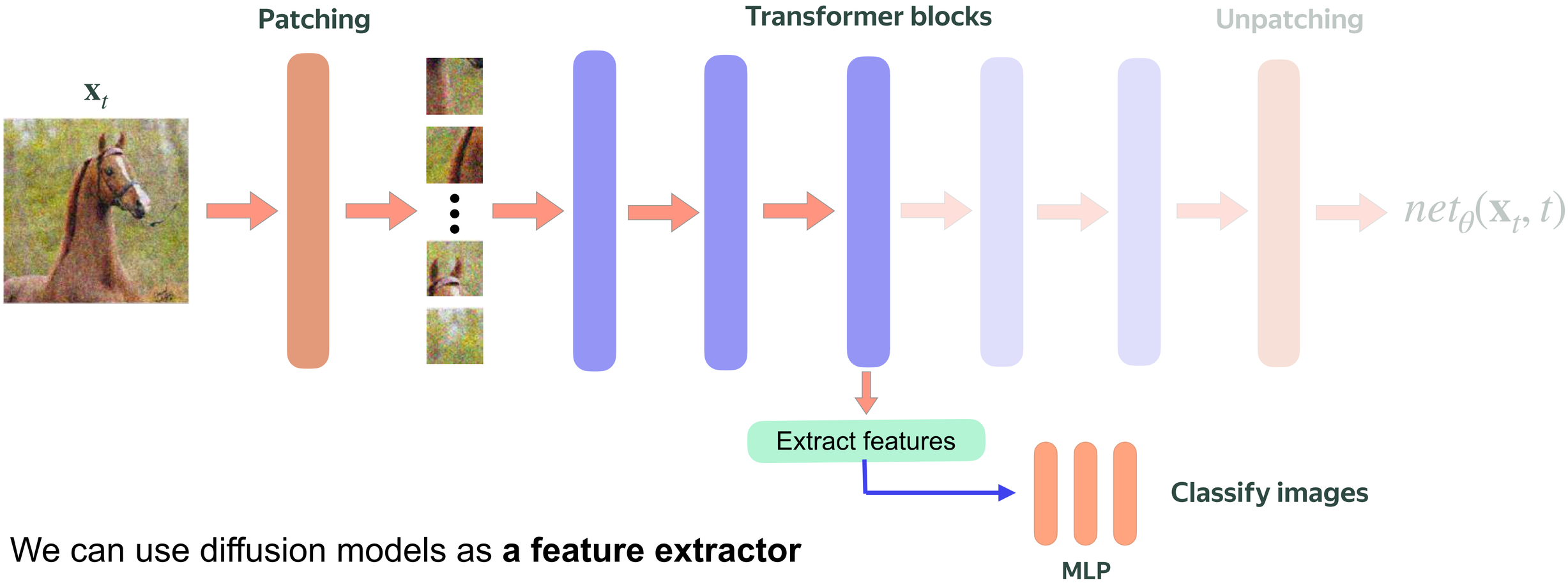
Diffusion model as representation learners

Diffusion models learn high quality image internal representations



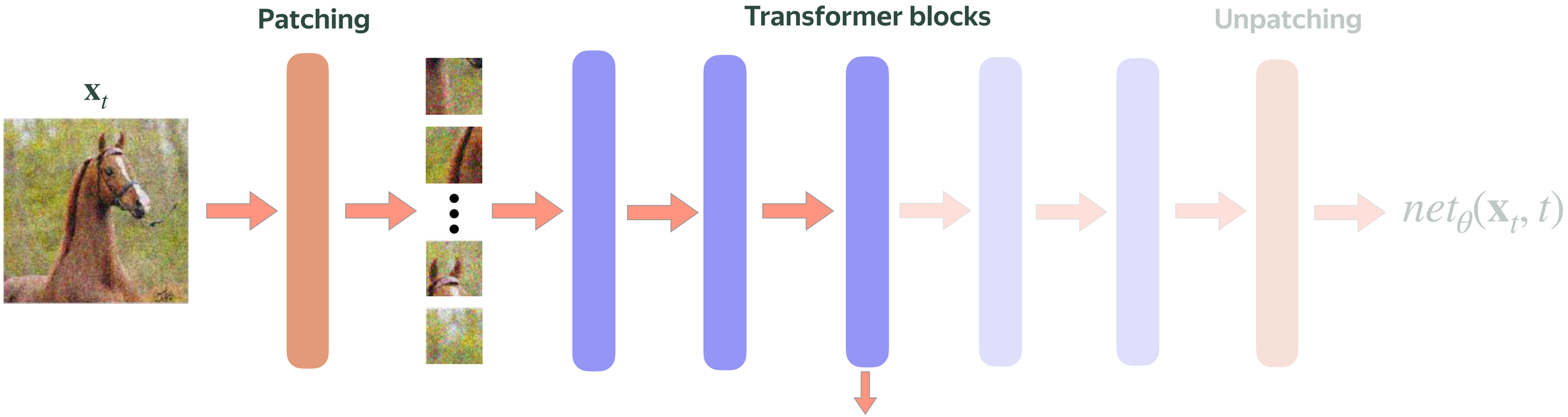
D. Baranchuk et al., Label-Efficient Semantic Segmentation with Diffusion Models, 2022

Diffusion model as representation learners



We can use diffusion models as **a feature extractor**

Diffusion model as representation learners

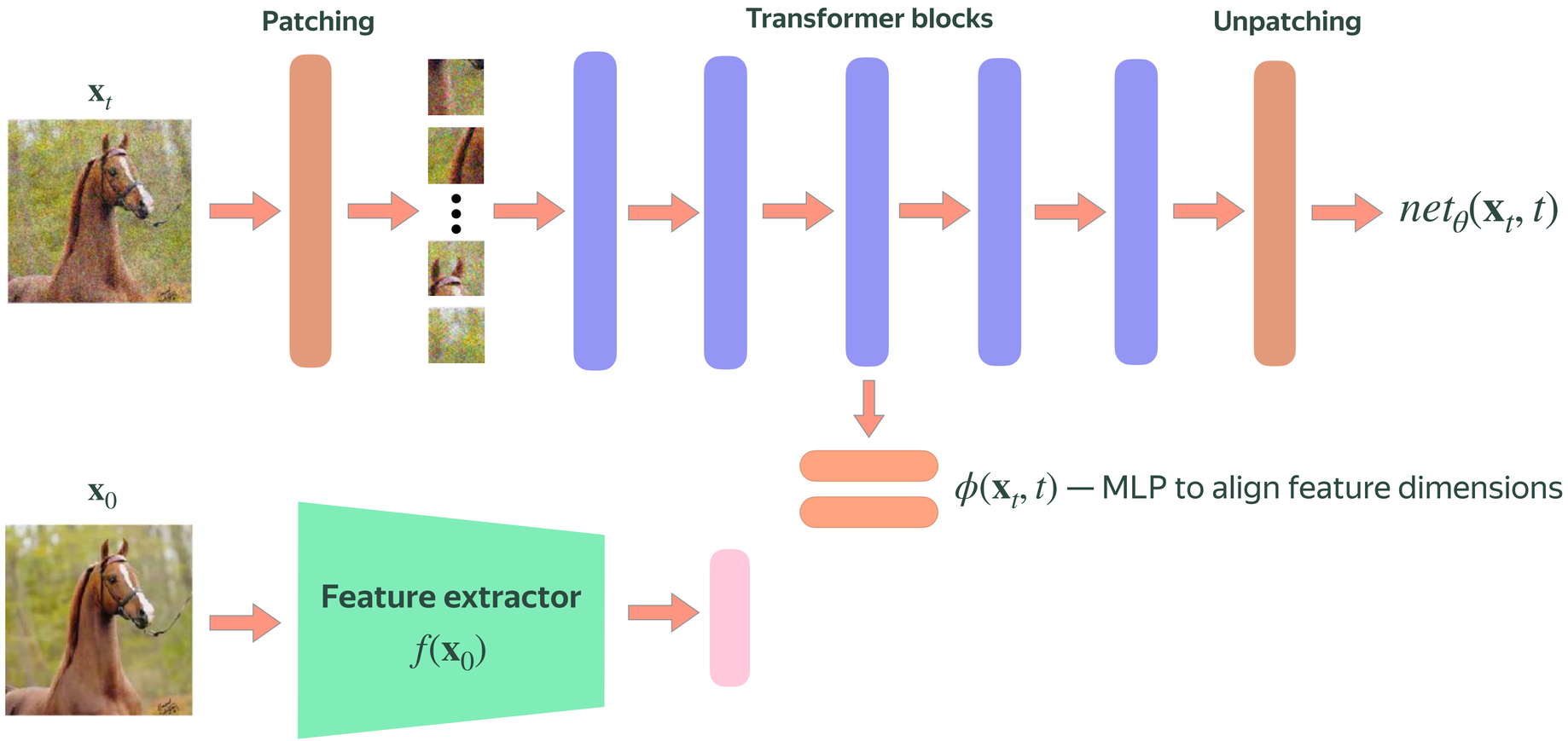


Is it difficult to learn good image representations? **Yes!**

Diffusion models spend much training time on representation learning

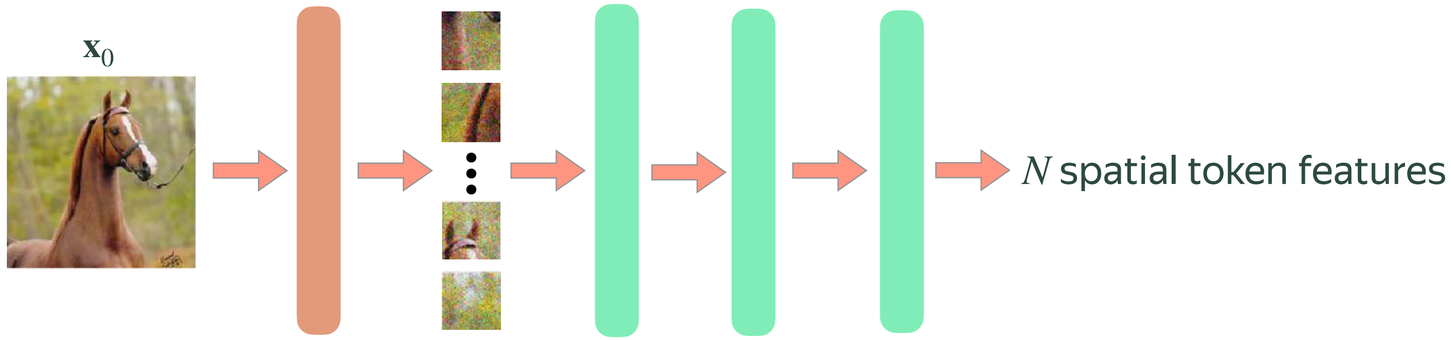
Representation Alignment | REPA

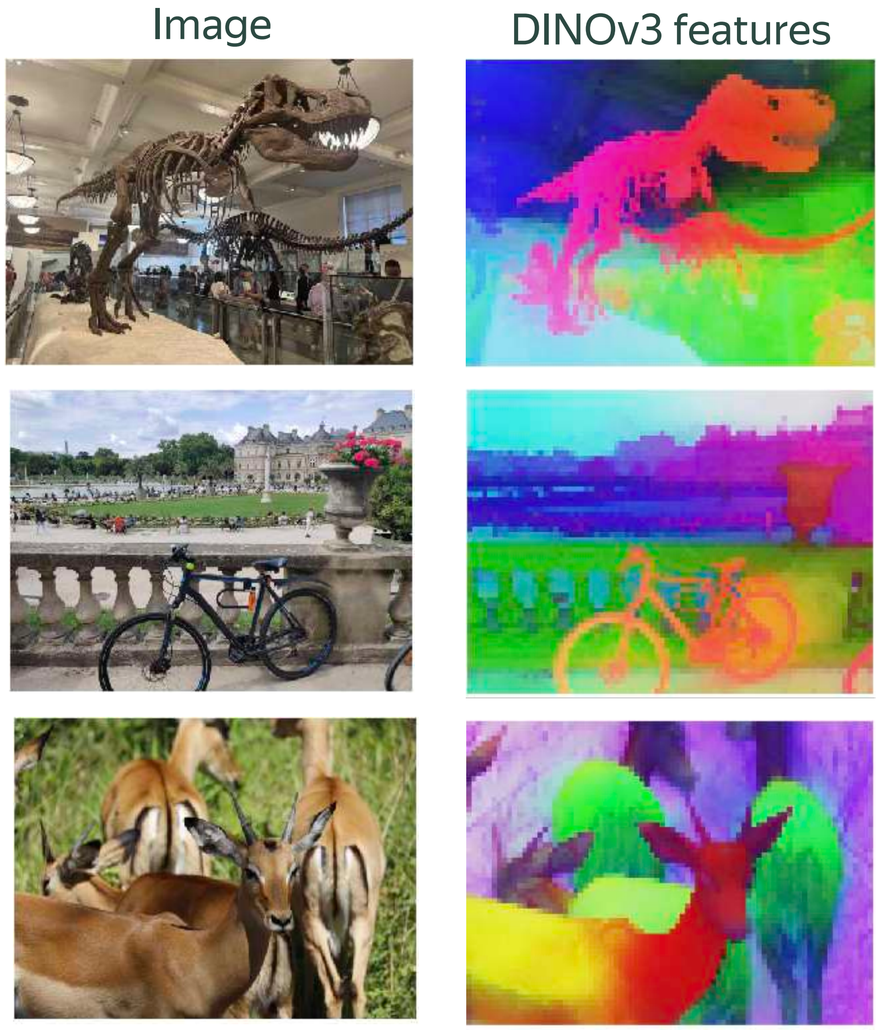
Idea: align diffusion features with a pretrained feature extractor



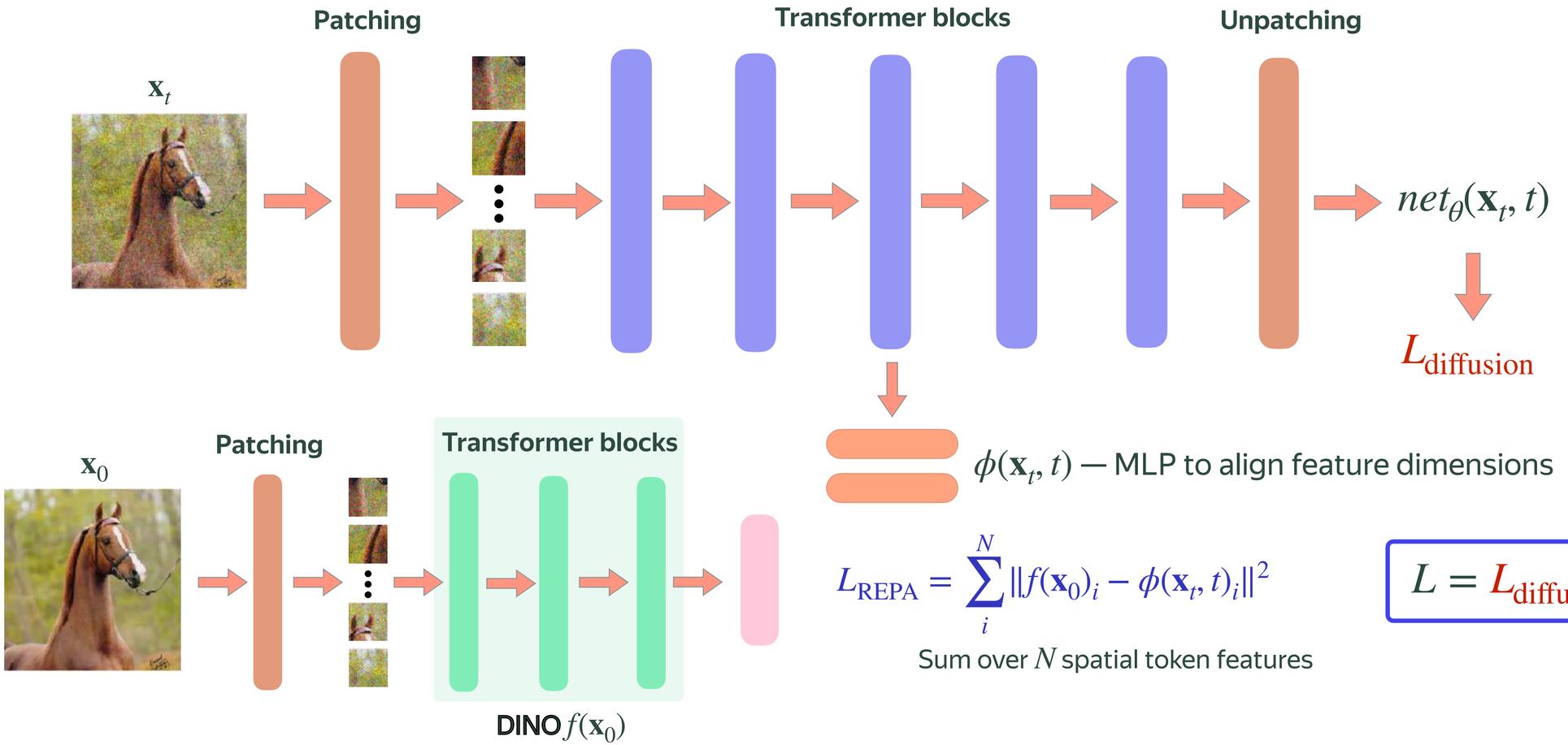
Pretrained feature extractor

DINOvX — state-of-the-art image feature extractors today



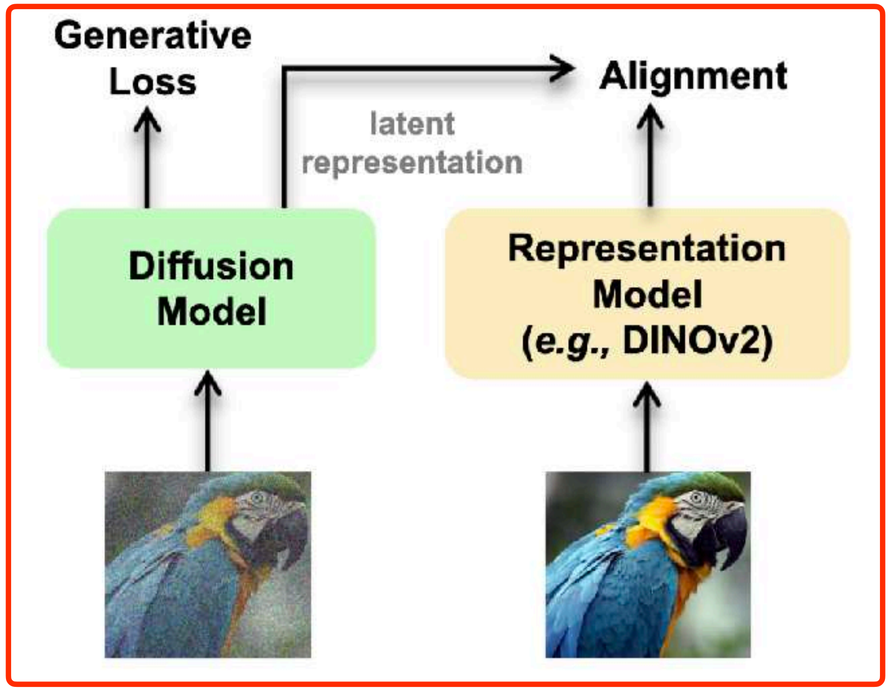


REPA with DINO



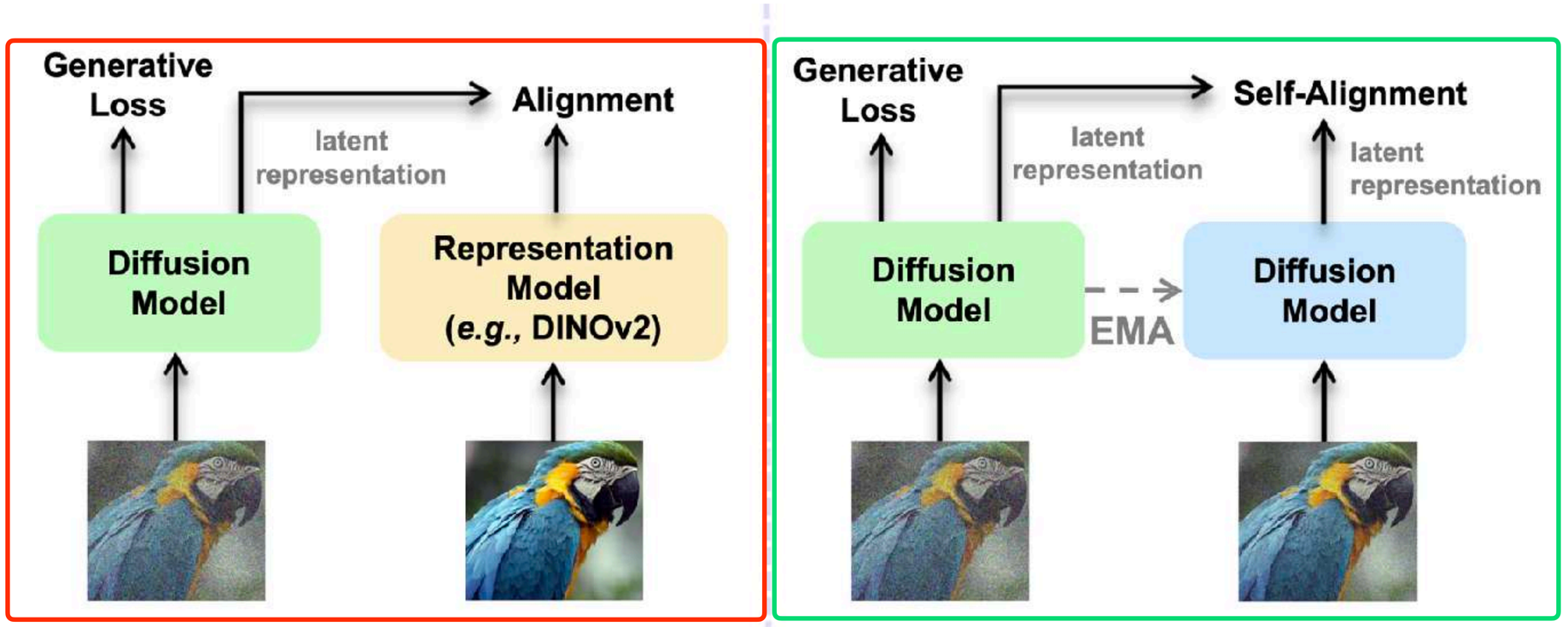
REPA problem

REPA with external encoders does not scale well for large text-to-image models



Self-alignment

Self-alignment: align diffusion features with those from its EMA version



D. Jiang et al., Diffusion Transformers Can Provide Representation Guidance by Themselves, 2025

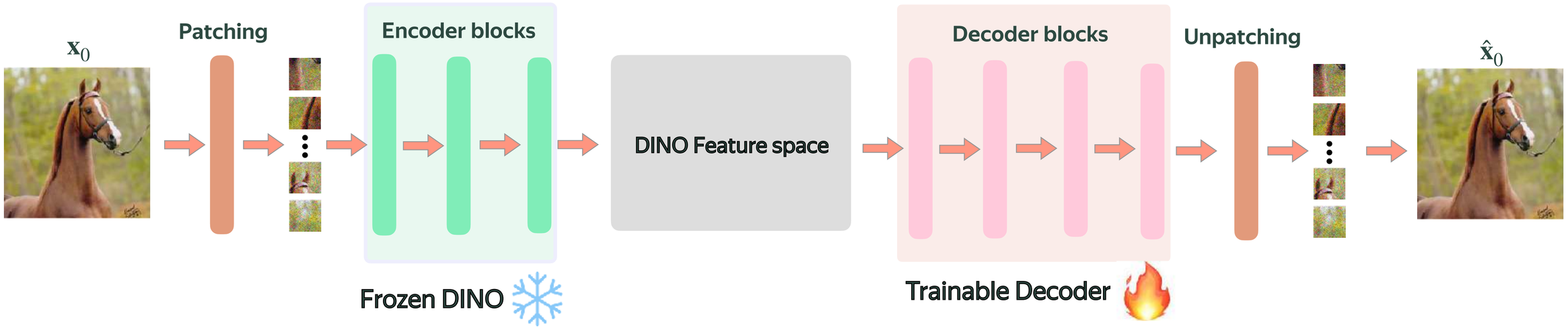
RAE | Representation autoencoder

If DINO helps diffusion training, maybe **train it directly in DINO feature space?**

B. Zheng et al., Diffusion Transformers with Representation Autoencoders, 2025

RAE | Training autoencoder

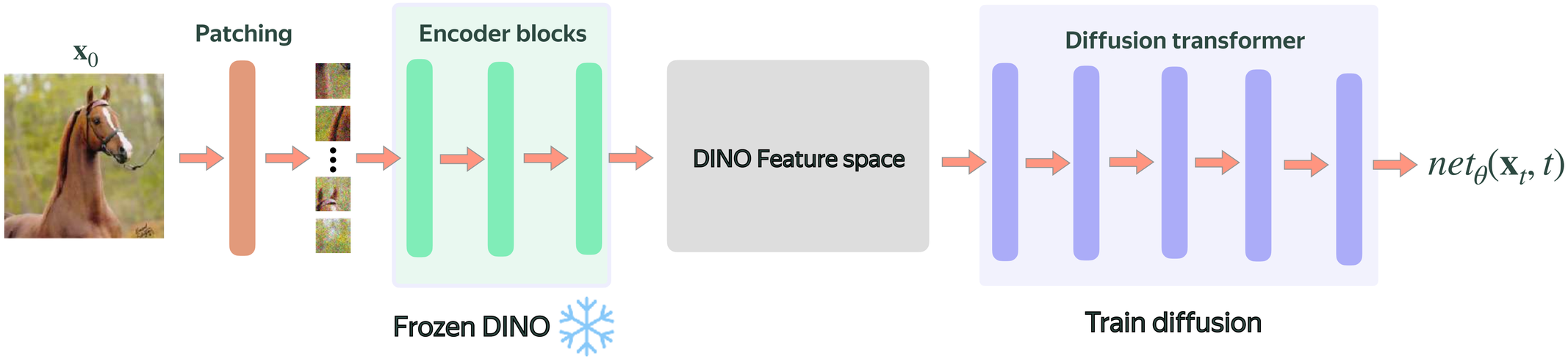
1) Train AE, where DINO is a frozen encoder



B. Zheng et al., Diffusion Transformers with Representation Autoencoders, 2025

RAE | Training diffusion

2) Train diffusion in DINO feature space



B. Zheng et al., Diffusion Transformers with Representation Autoencoders, 2025

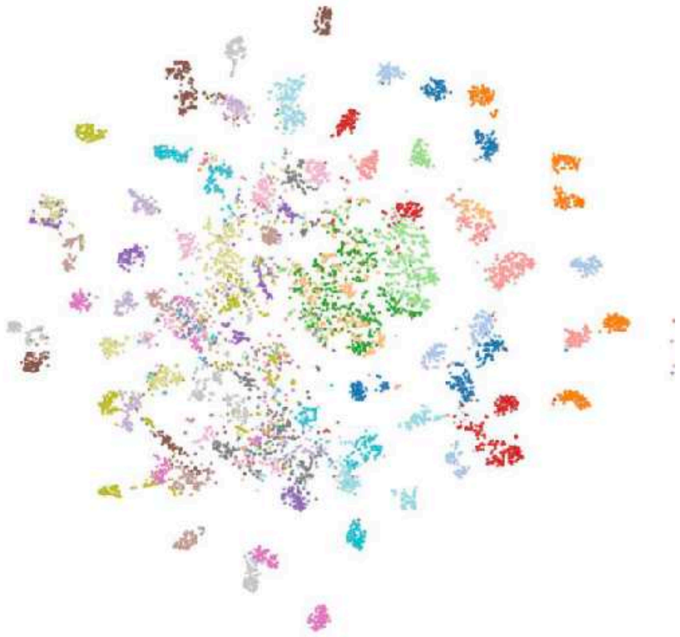
Why RAE?

DINO provides highly structured latent space

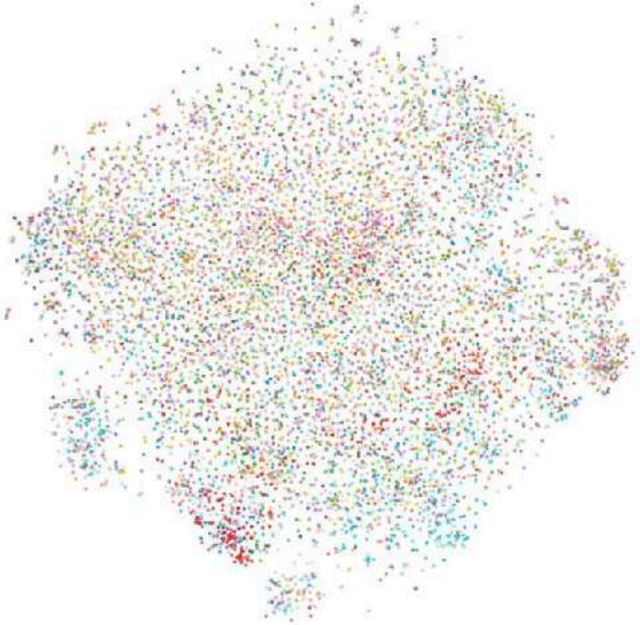
Very high learnability

Poor reconstruction quality
(Work in progress)

Latent space structures



DINOv3



SD-VAE

B. Zheng et al., Diffusion Transformers with Representation Autoencoders, 2025

Reconstruction examples

RAE has problems with low-level (colors) and fine-grained details

Takeaways

Diffusion models learn strong semantic representations in addition to generation

Diffusion models spend much training time to learning internal feature representations

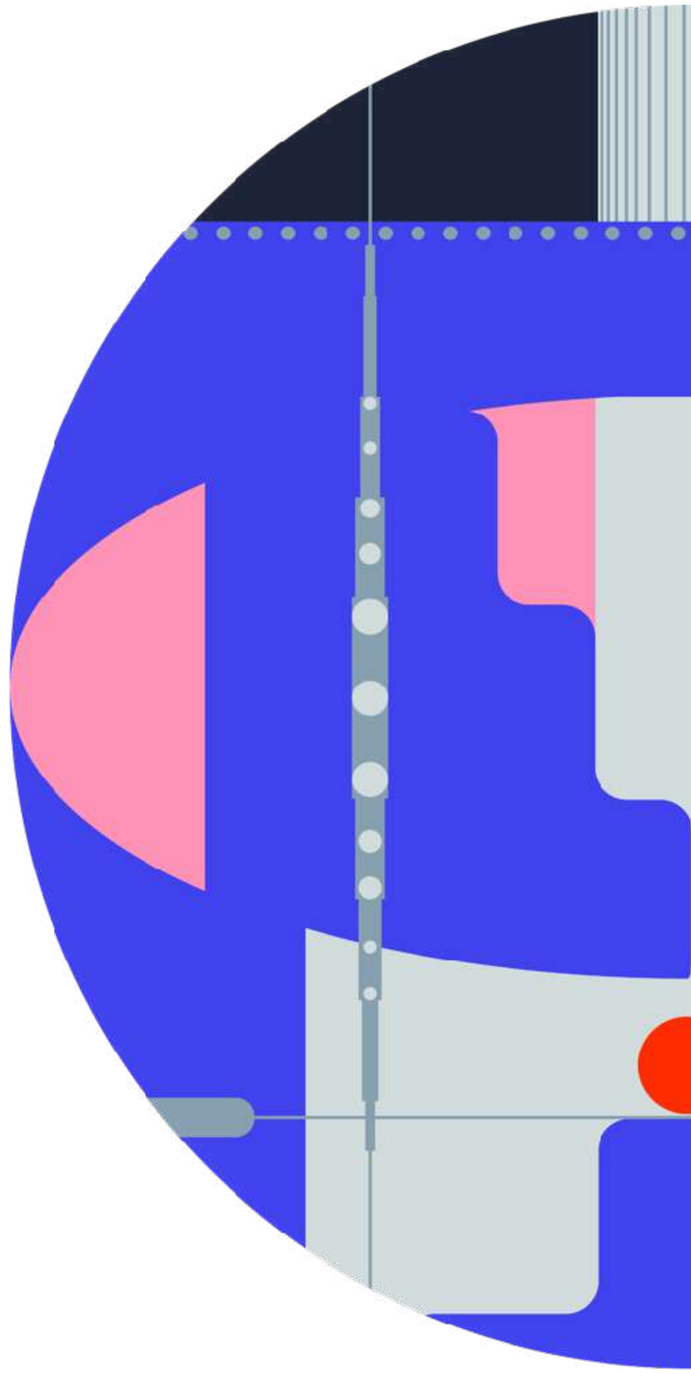
REPA improves and accelerates diffusion feature learning

In s.o.t.a. T2I, REPA + external encoders works poorly. Self-aligned diffusion features are promising

RAE enables training diffusion directly in semantic latent space

RAE offers great learnability but poor reconstruction; the direction is actively developing

How to sample with diffusion models?



Sampling schedule

$$s(\alpha, \cdot) : t \mapsto \frac{\alpha t}{1 + (\alpha - 1)t}$$

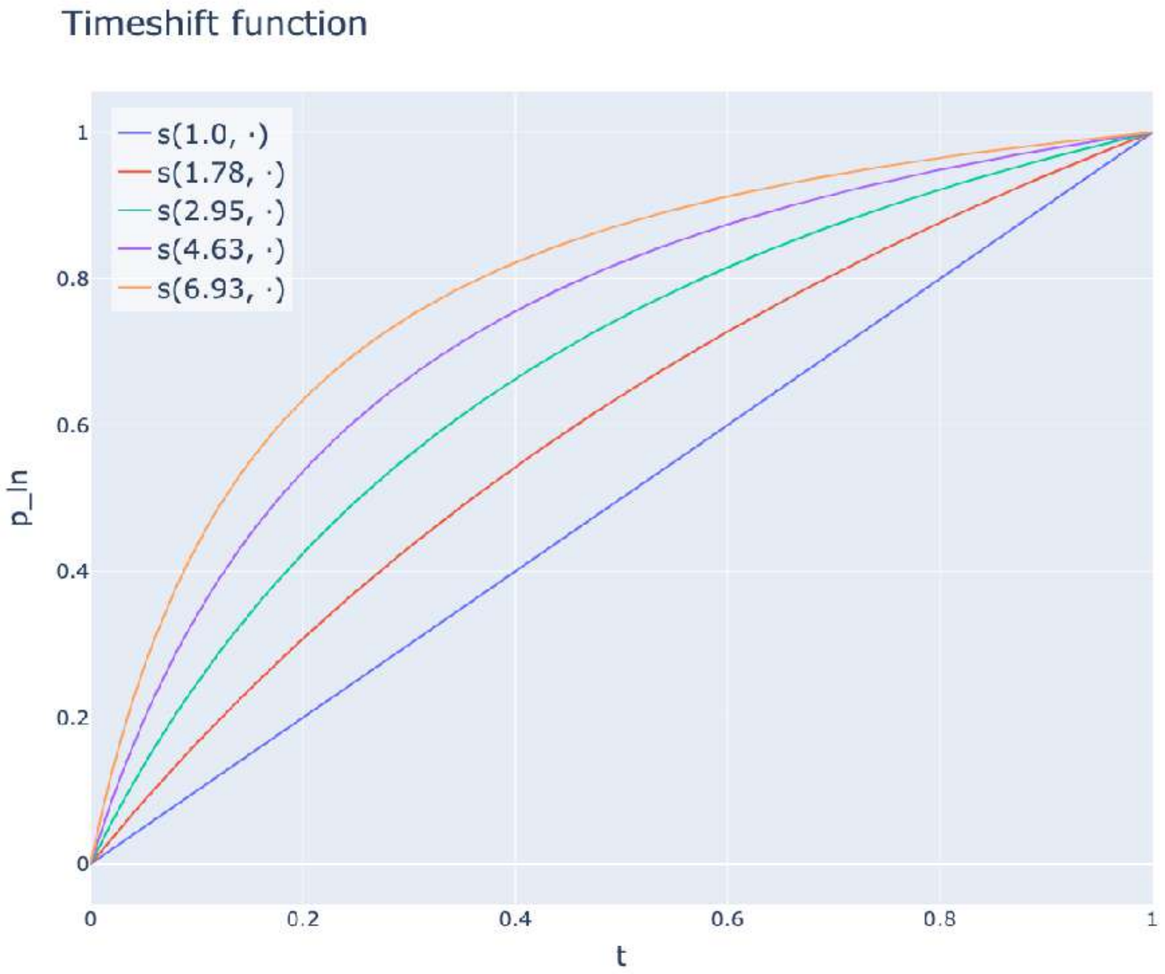
Similar intuition as at training

↓

Higher resolution → larger shift α

$\alpha = 1$ — uniform sampling

Okay in low-dimensional spaces



FLUX.2: Analyzing and Enhancing the Latent Space of FLUX -- Representation Comparison, 2025

Why this specific $s(\alpha, \cdot\,)$?

Why exactly this $s(\alpha, \cdot\,) : t \mapsto \dfrac{\alpha t}{1 + (\alpha - 1)t}$?

Why this specific $s(\alpha, \cdot)$?

Recall our dummy variable $Y(t) = (1-t)c + t\epsilon,$

$$Var\left[Y(t)\right] = \frac{t^2}{(1-t)^2}\frac{1}{N} \rightarrow \sigma(t, N) = \frac{t}{1-t}\sqrt{\frac{1}{N}}$$

Let's fix N and increase image size from N to M

What is t_{new} to make $\sigma(t, N) = \sigma(t_{new}, M)$?

Why this specific $s(\alpha, \cdot)$?

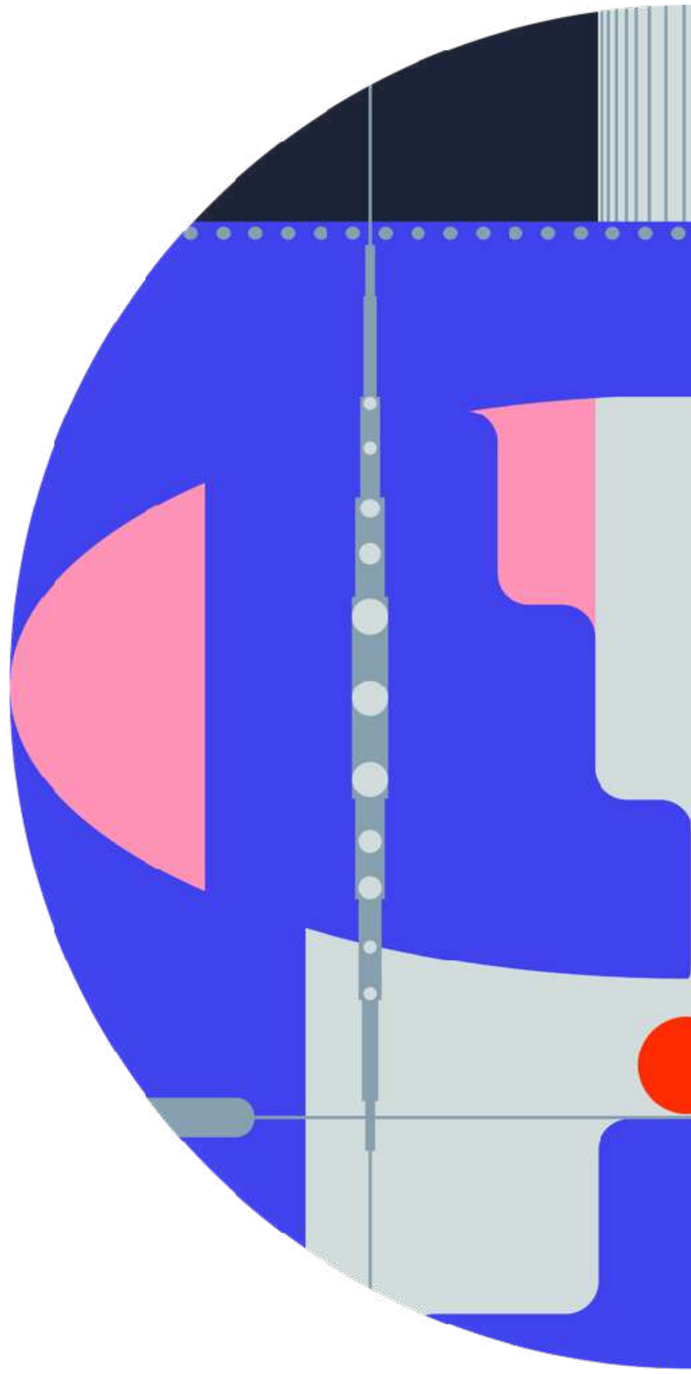
$$\sigma(t_{new}, M) = \sigma(t, N) \rightarrow \frac{t_{new}}{1 - t_{new}}\sqrt{\frac{1}{M}} = \frac{t}{1 - t}\sqrt{\frac{1}{N}}$$

$$\frac{1}{1 - t_{new}} - 1 = \frac{t}{1 - t}\sqrt{\frac{M}{N}} \quad \Rightarrow \quad 1 - t_{new} = \frac{1}{1 + \frac{t}{1 - t}\sqrt{\frac{M}{N}}} \quad \Rightarrow \quad t_{new} = \frac{\frac{t}{1 - t}\sqrt{\frac{M}{N}}}{1 + \frac{t}{1 - t}\sqrt{\frac{M}{N}}}$$

$$t_{new} = \frac{t\sqrt{\frac{M}{N}}}{1 + t(\sqrt{\frac{M}{N}} - 1)} = \frac{\alpha t}{1 + (\alpha - 1)t}, \quad \text{where } \alpha = \sqrt{\frac{M}{N}}$$

α controls how the schedule shifts when scaling resolution $N \rightarrow M$ from a uniform optimal schedule

Advanced guidance techniques



Classifier-free Guidance

CFG is critical for all practical generation scenarios

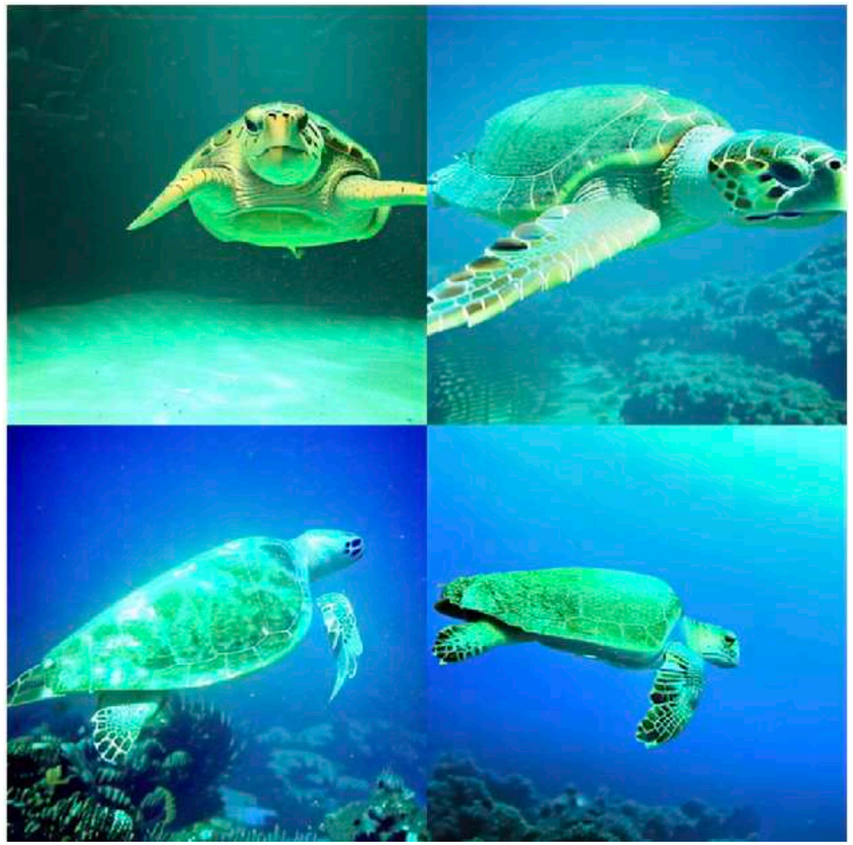
$$\mathbf{v}_{CFG}(\mathbf{x} \mid \mathbf{c}) = \mathbf{v}_\theta(\mathbf{x}, \varnothing) + w \cdot (\mathbf{v}_\theta(\mathbf{x}, \mathbf{c}) - \mathbf{v}_\theta(\mathbf{x}, \varnothing)), \;\; w \geq 1$$

Improves condition alignment and quality

Reduces diversity and oversaturates images for large guidance scales

High CFG problems

Low diversity & Oversaturation

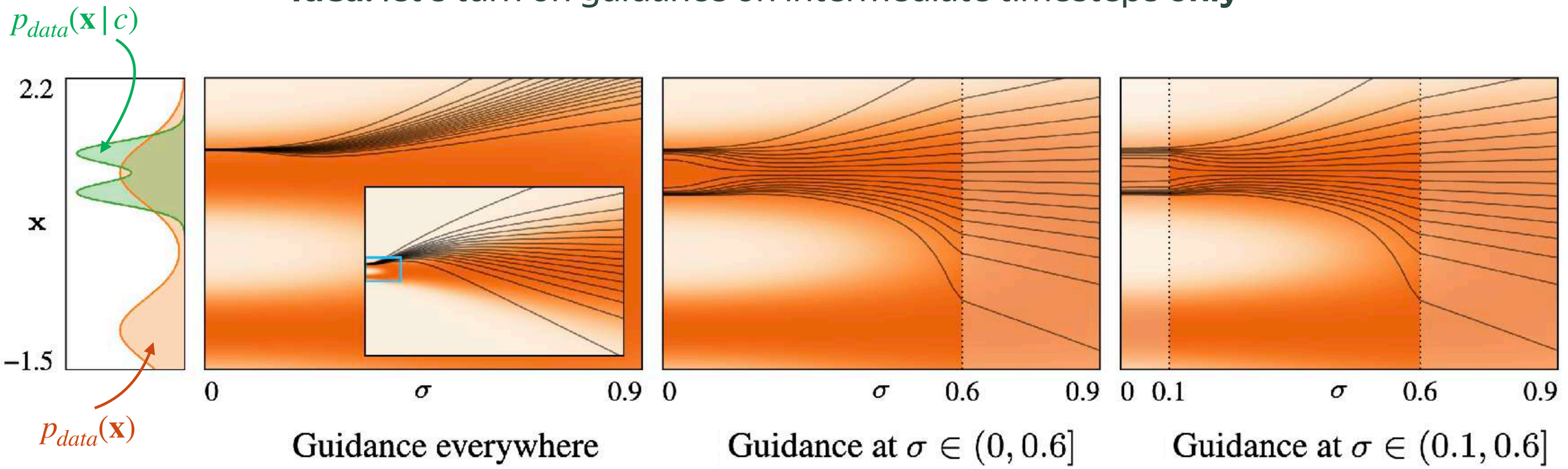


Oversaturation

T. Kynkäänniemi et al., Applying Guidance in a Limited Interval Improves Sample and Distribution Quality in Diffusion Models, 2024

Interval Guidance

Idea: let's turn on guidance on intermediate timesteps **only**



Miss one mode

Do not miss modes

Nothing changes if turn off at the end

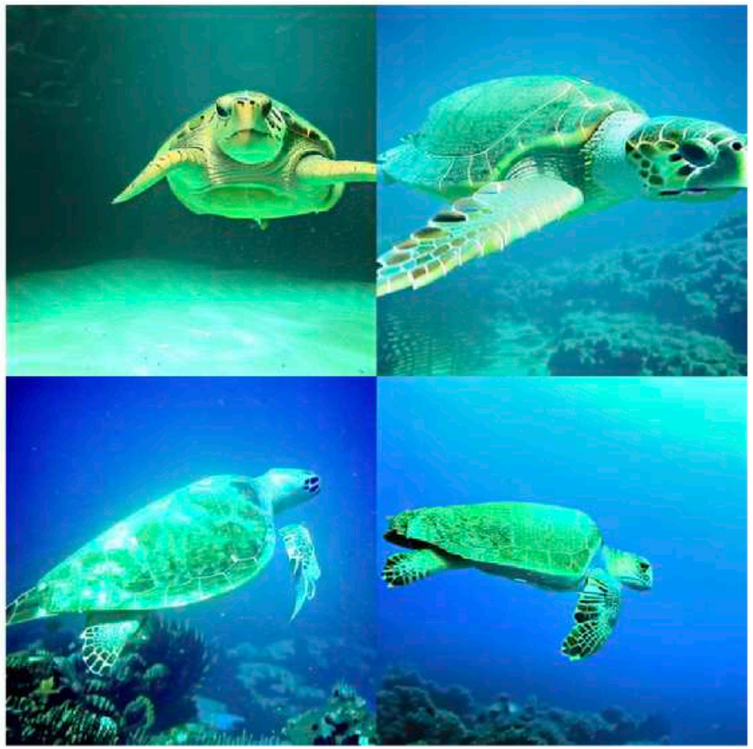
T. Kynkäänniemi et al., Applying Guidance in a Limited Interval Improves Sample and Distribution Quality in Diffusion Models, 2024

Interval Guidance

T. Kynkäänniemi et al., Applying Guidance in a Limited Interval Improves Sample and Distribution Quality in Diffusion Models, 2024

Interval Guidance

No CFG

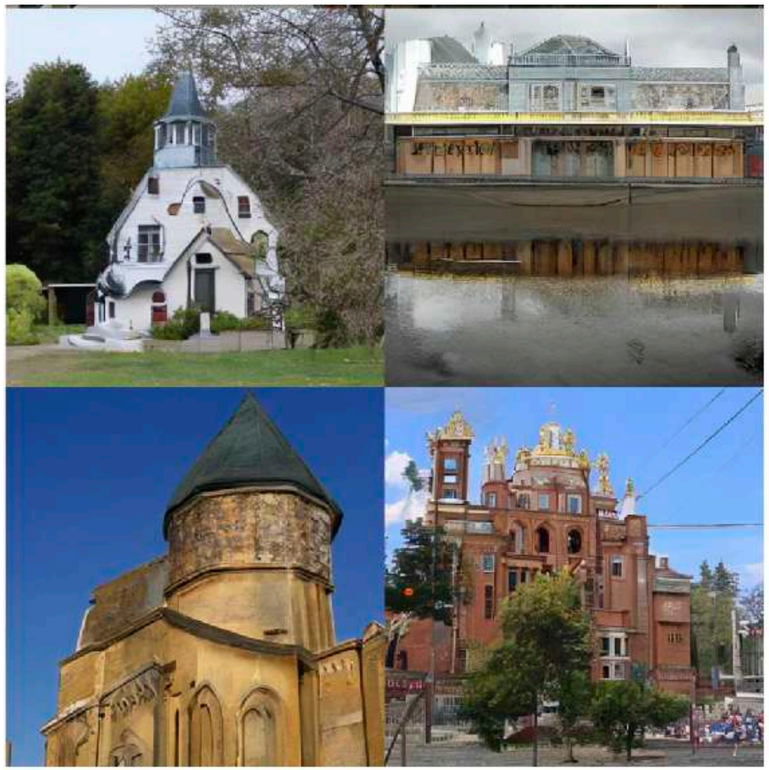


High constant CFG

Interval CFG

T. Kynkäänniemi et al., Applying Guidance in a Limited Interval Improves Sample and Distribution Quality in Diffusion Models, 2024

Interval Guidance



No CFG

High constant CFG

Interval CFG

T. Kynkäänniemi et al., Applying Guidance in a Limited Interval Improves Sample and Distribution Quality in Diffusion Models, 2024

CFG with negative conditioning

We can replace $\mathbf{v}_\theta(\mathbf{x}, \varnothing)$ with **anything** we do not want to generate

For example, we can use $\mathbf{v}_\theta(\mathbf{x}, \mathbf{c}^-)$, where $\mathbf{c}^-$ is some negative condition

$\mathbf{c}^+$ = “Garfield”, $\mathbf{c}^-$ = “”

$\mathbf{c}^+$ = “Garfield”, $\mathbf{c}^-$ = “a cat”

CFG with negative conditioning

Negative prompt $\mathbf{c}^-$ example in practice

“low quality, blurry, out of focus, low resolution, pixelated, jpeg artifacts, compression artifacts, noisy, grainy, oversharpened, aliasing, banding, bad anatomy, bad proportions, deformed, disfigured, malformed limbs, mutated hands, extra fingers, missing fingers, fused fingers, extra limbs, long neck, twisted body, duplicate body parts, poorly drawn face, distorted face, asymmetrical eyes, cross-eyed, bad eyes, dead eyes, distorted pupils cropped, out of frame, cut off limbs, poorly framed, bad composition, tilted horizon watermark, 3d render, plastic skin, waxy skin, doll-like, overexposed, underexposed, harsh shadows, color banding, color bleeding, duplicate, cloned face”

AutoGuidance

Let's consider a bad version of $\mathbf{v}_\theta(\mathbf{x}, \mathbf{c})$, e.g., *early checkpoint or smaller model*

$\mathbf{v}_\theta^{bad}(\mathbf{x}, \mathbf{c})$ is trained on the *same conditioning, same distribution*

but slightly worse than the main model $\mathbf{v}_\theta^{main}(\mathbf{x}, \mathbf{c})$

$$\mathbf{v}_{AG}(\mathbf{x} \mid \mathbf{c}) = \mathbf{v}_\theta^{bad}(\mathbf{x}, \mathbf{c}) + w \cdot (\mathbf{v}_\theta^{main}(\mathbf{x}, \mathbf{c}) - \mathbf{v}_\theta^{bad}(\mathbf{x}, \mathbf{c})), \quad w \geq 1$$

T. Karras et al., Guiding a Diffusion Model with a Bad Version of Itself, 2024

Why AG works?

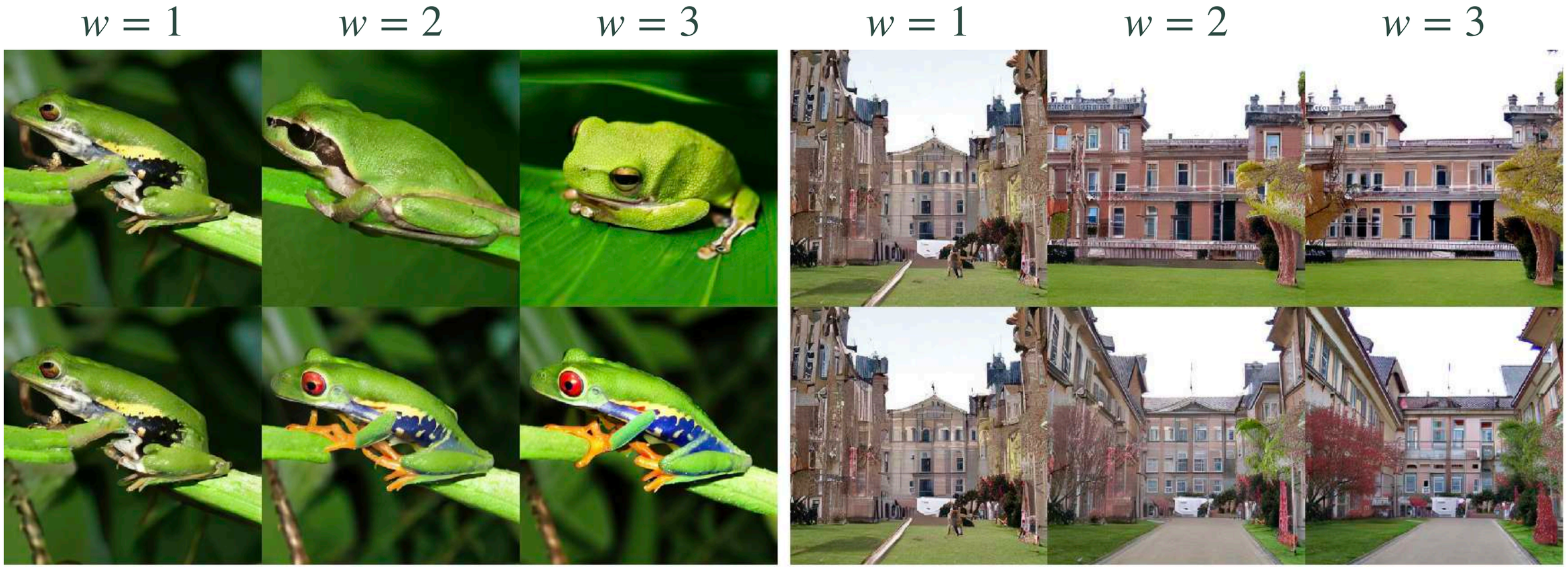
$\mathbf{v}_\theta^{bad}(\mathbf{x}, \mathbf{c})$ is **globally aligned** with $\mathbf{v}_\theta^{main}(\mathbf{x}, \mathbf{c})$ but $\mathbf{v}_\theta^{bad}(\mathbf{x}, \mathbf{c})$ points to more "**error**" directions

Conceptual decomposition: $\mathbf{v}_\theta^{bad}(\mathbf{x}, \mathbf{c}) = \mathbf{v}_\theta^{main}(\mathbf{x}, \mathbf{c}) + \delta \cdot \mathbf{v}_\theta^{error}(\mathbf{x}, \mathbf{c})$

$$\mathbf{v}_{AG}(\mathbf{x} \mid \mathbf{c}) = \mathbf{v}_\theta^{bad}(\mathbf{x}, \mathbf{c}) + w \cdot (\mathbf{v}_\theta^{main}(\mathbf{x}, \mathbf{c}) - \mathbf{v}_\theta^{bad}(\mathbf{x}, \mathbf{c})) = \boxed{\mathbf{v}_\theta^{main}(\mathbf{x}, \mathbf{c}) - \hat{w} \cdot \mathbf{v}_\theta^{error}(\mathbf{x}, \mathbf{c})}$$

T. Karras et al., Guiding a Diffusion Model with a Bad Version of Itself, 2024

Results



T. Karras et al., Guiding a Diffusion Model with a Bad Version of Itself, 2024

Takeaways

Guidance — a **core** technique in **all** diffusion models in practice

Classifier-Free Guidance — **default** for text-to-image / video. Often, **negative prompt** instead of ∅

Interval Guidance — simple and popular idea to increase diversity and reduce oversaturation

AutoGuidance — highly promising idea but hard to find a proper guiding (“bad”) model in practice

There are **numerous** variants of guidance approaches

Different guidance methods can be combined together